

Finding Small Steps for Hard Questions, in work on the Process Semantics of Regular Expressions

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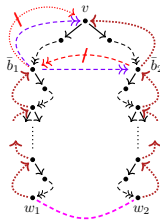
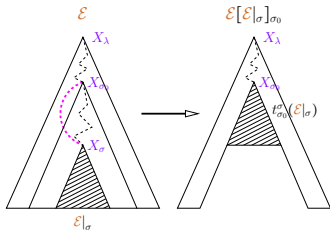


L'Aquila, Italy

IFIP 1.6 Working Group Meeting

Lisbon

July 18, 2026



Overview

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) reduction steps for well-behaved 1-graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, as far as possible
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback **correspondences**
 - (irc) LLEE/1-LLEE graphs $\triangleq$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

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Regular Expression and Milner's Process Interpretation

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Process interpretation P of regular expressions *(Milner, 1984)*

$0 \xrightarrow{P} \text{deadlock } \delta, \text{ no termination}$

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Regular Expressions

Definition (*~ Copi-Elgot-Wright, 1958*)

Regular expressions over alphabet A with unary Kleene star:

$e, e_1, e_2 ::= 0 \mid a \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e^* \quad (\text{for } a \in A).$

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- ▶ with unary star * : **1** is definable as **0^{*}**

Regular Expressions

Definition ($\sim$ Kleene, 1951, $\sim$ Copi-Elgot-Wright, 1958)

Regular expressions over alphabet A with unary / binary Kleene star:

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$$e, e_1, e_2 ::= 0 \mid 1 \mid a \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e_1^{\oplus} e_2 \quad (\text{for } a \in A).$$

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Regular Expressions (1-free)

Definition ($\sim$ Kleene, 1951, $\sim$ Copi-Elgot-Wright, 1958)

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Definition (for process interpretation)

1-free regular expressions over alphabet A with binary Kleene star:

$$f, f_1, f_2 ::= 0 \mid a \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid f_1^{\oplus} f_2 \quad (\text{for } a \in A).$$

Regular Expressions (1-free)

Definition ($\sim$ Kleene, 1951, $\sim$ Copi-Elgot-Wright, 1958)

Regular expressions over alphabet A with unary / binary Kleene star:

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Definition (for process interpretation)

1-free regular expressions over alphabet A with unary / binary Kleene star:

$$f, f_1, f_2 ::= 0 \mid a \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid (f_1^*) \cdot f_2 \quad (\text{for } a \in A),$$

$$f, f_1, f_2 ::= 0 \mid a \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid f_1^{\oplus} f_2 \quad (\text{for } a \in A).$$

Regular Expressions (under-star-/1-free)

Definition ($\sim$ Kleene, 1951, $\sim$ Copi-Elgot-Wright, 1958)

Regular expressions over alphabet A with unary / binary Kleene star:

$$e, e_1, e_2 ::= \mathbf{0} \mid \mathbf{1} \mid a \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e^* \quad (\text{for } a \in A).$$

$$e, e_1, e_2 ::= \mathbf{0} \mid \mathbf{1} \mid a \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e_1^{\oplus} e_2 \quad (\text{for } a \in A).$$

- ▶ symbol $\mathbf{0}$ instead of $\emptyset$, symbol $\mathbf{1}$ instead of $\{\epsilon\}$
- ▶ with unary star * : $\mathbf{1}$ is definable as $\mathbf{0}^*$
- ▶ with binary star $^{\oplus}$: $\mathbf{1}$ is **not** definable (in its absence)

Definition (for process interpretation)

The set $RExp^{(+)}(A)$ of **1-free** regular expressions over A is defined by:

$$f, f_1, f_2 ::=_{\text{assoc}(\cdot)} \mathbf{0} \mid a \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid f_1^* \cdot f_2 \quad (\text{for } a \in A),$$

Regular Expressions $((*/\perp)/1\text{-free})$

Definition ($\sim$ Kleene, 1951, $\sim$ Copi-Elgot-Wright, 1958)

Regular expressions over alphabet A with unary / binary Kleene star:

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the set $RExp^{(*\perp)}(A)$ of **under-star-1-free $((*/\perp)$ reg. expr.** over A by:

$$uf, uf_1, uf_2 ::= 0 \mid 1 \mid a \mid uf_1 + uf_2 \mid uf_1 \cdot uf_2 \mid f^* \quad (\text{for } a \in A).$$

Process semantics $\llbracket \cdot \rrbracket_P$ of regular expressions (Milner, 1984)

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$1 \xrightarrow{P}$ empty-step process ϵ , then terminate

$a \xrightarrow{P}$ atomic action a , then terminate

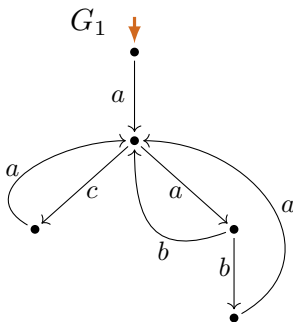
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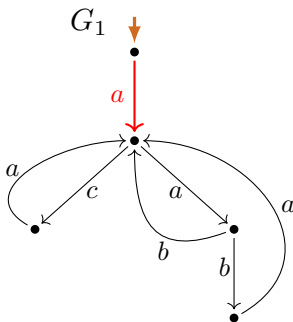
$\llbracket e \rrbracket_P := \llbracket P(e) \rrbracket_{\leftrightarrow}$ (bisimilarity equivalence class of process $P(e)$)

P -expressibility and $[[\cdot]]_P$ -expressibility (example, informally)



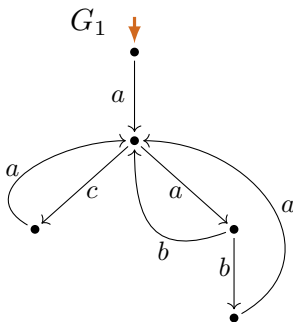
$$P\left(\overbrace{a \cdot (c \cdot a + a \cdot (b + b \cdot a))^* \cdot 0}^f \right)$$

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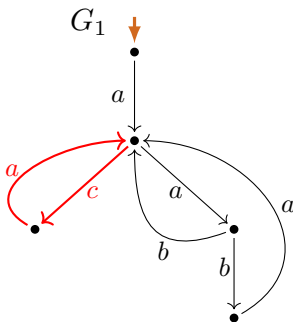
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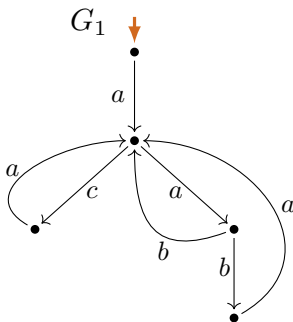
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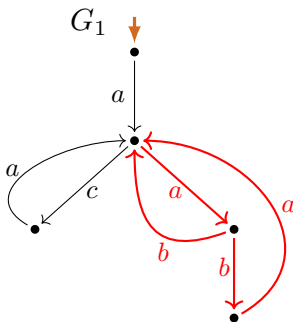
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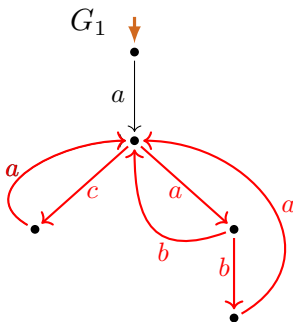
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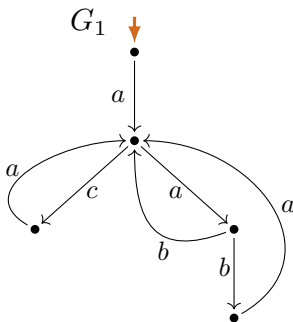
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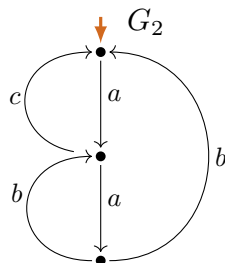
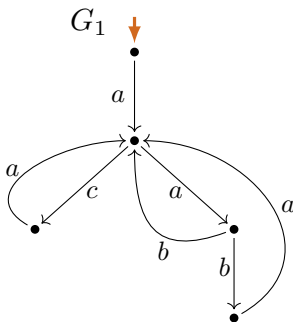
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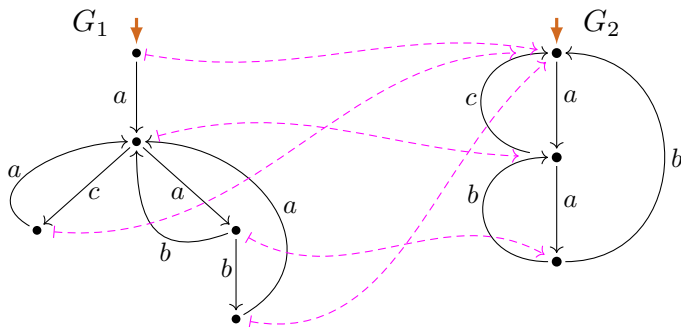
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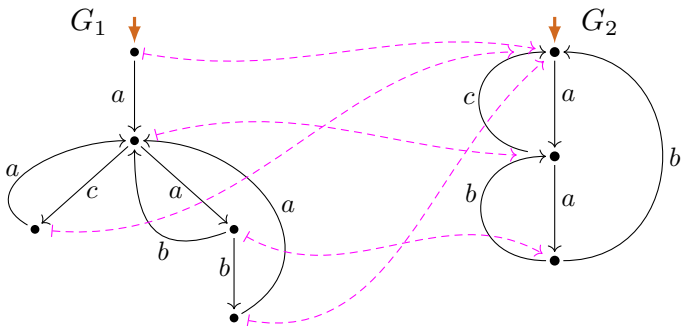
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$$G_1 \in \llbracket f \rrbracket_P$$

$$G_2 \in \llbracket f \rrbracket_P$$

Process interpretation P (formally)

Definition (Transition system specification $\mathcal{T}$)

$$\frac{}{a \xrightarrow{a} 1} \quad \frac{e_i \xrightarrow{a} e'_i}{e_1 + e_2 \xrightarrow{a} e'_i} \quad (i \in \{1, 2\})$$

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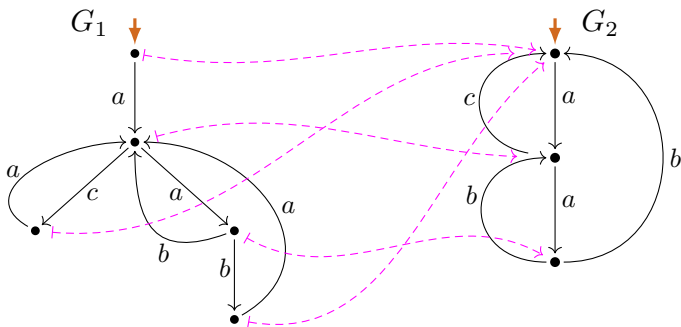
Definition

The **process interpretation** $P(e)$ of a regular expression e :

$P(e) :=$ **labeled transition graph** generated by e by derivations in $\mathcal{T}$,

The **process semantics** $\llbracket e \rrbracket_P$ of e is: $\llbracket e \rrbracket_P := [P(e)]_{\leftrightarrow}$.

P -expressibility and $[[\cdot]]_P$ -expressibility (example, **informally**)

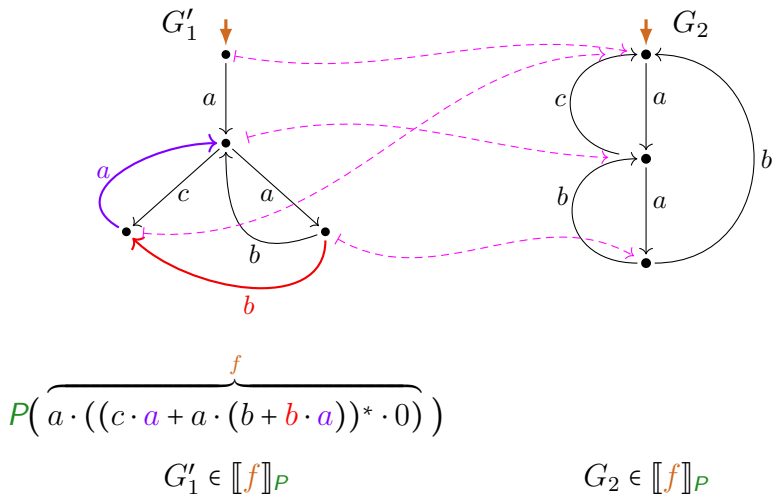


$$P\left(\overbrace{a \cdot ((c \cdot a + a \cdot (b + b \cdot a))^* \cdot 0))}^f\right)$$

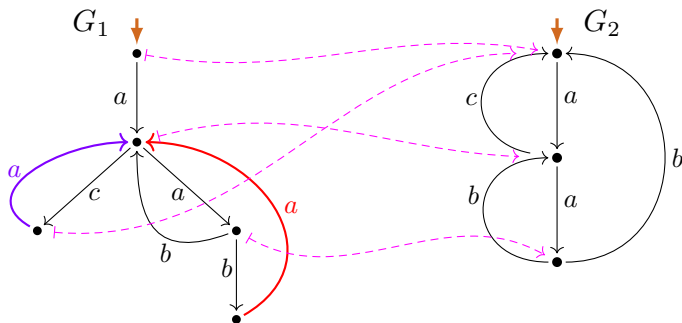
$$G_1 \in [[f]]_P$$

$$G_2 \in [[f]]_P$$

P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (example, formally)



P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (example, formally)

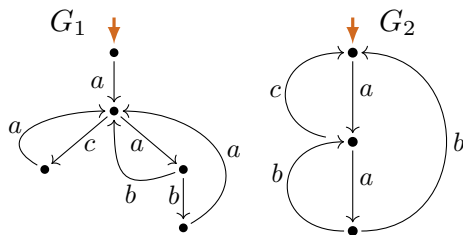


$$P\left(\overbrace{a \cdot ((c \cdot a + a \cdot (b + b \cdot (a + a)))^* \cdot 0)}^f\right)$$

$$G_1 \in \llbracket f \rrbracket_P$$

$$G_2 \in \llbracket f \rrbracket_P$$

P -expressibility and $[[\cdot]]_P$ -expressibility (examples)

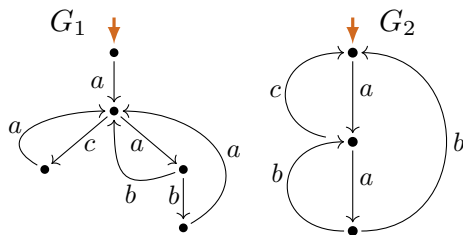


P -expressible

$[[\cdot]]_P$ -expressible

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P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (examples)



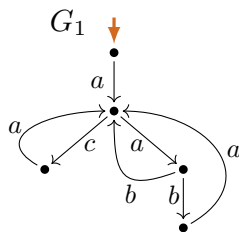
P -expressible

?

$\llbracket \cdot \rrbracket_P$ -expressible

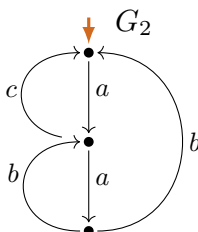
$\llbracket \cdot \rrbracket_P$ -expressible

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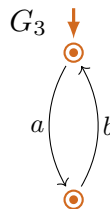
P -expressible

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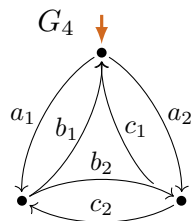
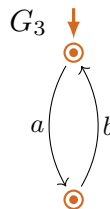
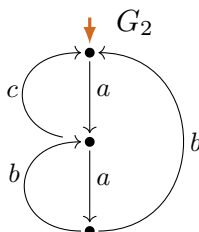
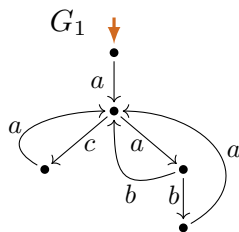
$[[\cdot]]_P$ -expressible



not P -expressible

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P -expressibility and $[[\cdot]]_P$ -expressibility (examples)



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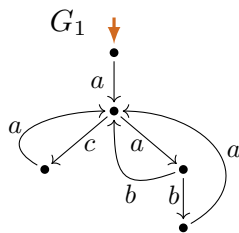
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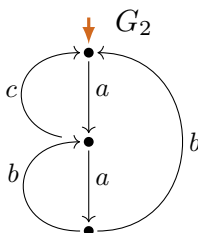
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P -expressibility and $[[\cdot]]_P$ -expressibility (examples)



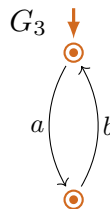
P -expressible

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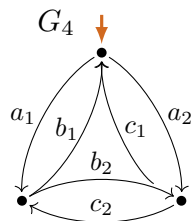
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$[[\cdot]]_P$ -expressible



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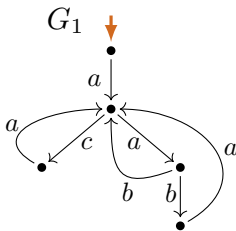
not $[[\cdot]]_P$ -expressible



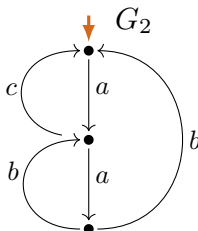
Expressibility Questions:

How can P -expressibility and $[[\cdot]]_P$ -expressibility be characterised?

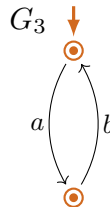
P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (examples)



P -expressible

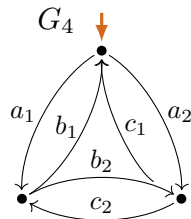
 $\llbracket \cdot \rrbracket_P$ -expressible

?

 $\|\cdot\|_P$ -expressible

not P -expressible

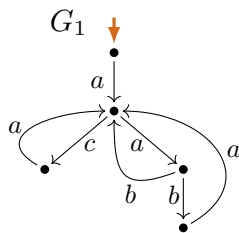
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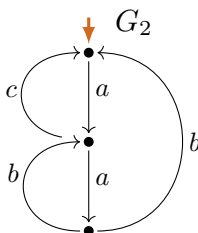
What structural property of finite charts [process graphs] is necessary and sufficient of star behaviour $[[\cdot]]_{p\text{-expressibility}}$ (Milner, 1984)

P -expressibility and $[[\cdot]]_P$ -expressibility (examples)



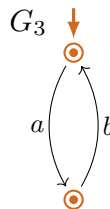
P -expressible

$[[\cdot]]_P$ -expressible



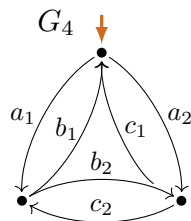
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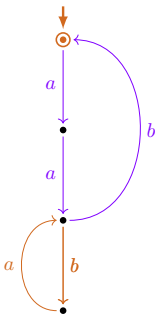
Overview

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) **reduction steps** for **well-behaved** 1-graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) **collapse steps** for **LLEE** graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) **collapse steps** for **LLEE** 1-graphs under bisimilarity, **as far as possible**
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback **correspondences**
 - (irc) **LLEE/1-LLEE** graphs $\hat{=}$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a **compact** refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

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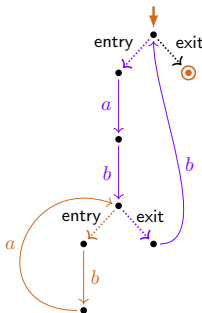
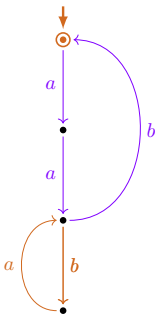
Well-behaved 1-graphs (with 1-transitions)



$$\sim P((aa(ba)^*b)^*)$$

$$P(1 \cdot (a \cdot ((a \cdot (b \cdot a)^*) \cdot b))^*)$$

Well-behaved 1-graphs (with 1-transitions)



well-behaved
loop-entry/-exit palm tree
(Corradini, Baeten)

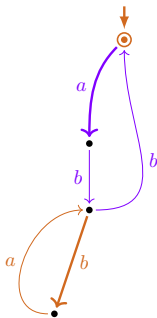
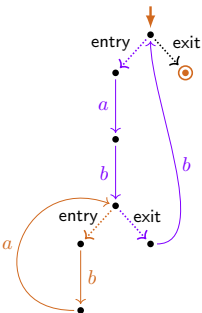
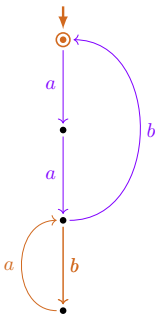
$$\sim P((aa(ba)^*b)^*)$$

$$\sim P((1 \cdot aa(1 \cdot ba)^* 1 \cdot b)^* (1 \cdot 1))$$

$$P(1 \cdot (a \cdot ((a \cdot (b \cdot a)^* \cdot b))^*))$$

$$P(1 \cdot (((1 \cdot a) \cdot (((a \cdot ((1 \cdot b) \cdot a)^* \cdot 1) \cdot b))^*) \cdot 1))$$

Well-behaved 1-graphs (with 1-transitions)



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$$\sim P((aa(ba)^*b)^*)$$

$$\sim P(((1aa(1ba)^*1b)^*(1.1)))$$

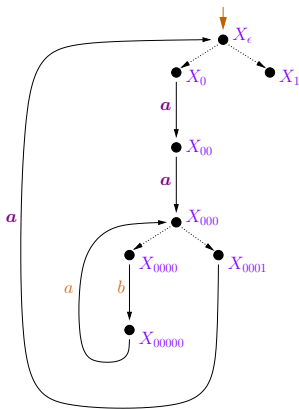
$$\sim P((aa(ba)^*b)^*)$$

$$P(1.(a.((a.(b.a)^*.b))^*)$$

$$P(1.((((1.a).(((a.((1.b).a)^*.1).b))^*.1)))$$

$$P(1.(a.((a.(b.a)^*.b))^*)$$

Well-Behaved specification (example)



$$X_\epsilon = 1 \cdot X_0 + 1 \cdot X_1$$

$$X_0 = a \cdot X_{00}$$

$$X_{00} = a \cdot X_{000}$$

$$X_{000} = 1 \cdot X_{0000} + 1 \cdot X_{0001}$$

$$X_{0000} = b \cdot X_{00000}$$

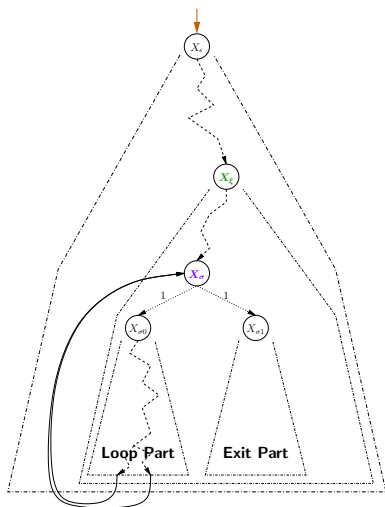
$$X_{00000} = a \cdot X_{000}$$

$$X_{0001} = a \cdot X_\epsilon$$

$$X_1 = 0$$

$$P((aa(ba)^*a)^*.0)$$

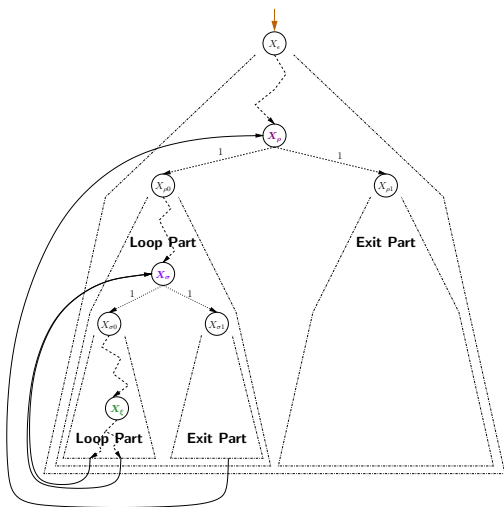
Well-Behaved specifications (well-beh./cycling variables)



$X_\xi, X_\epsilon \dots$ *well-behaved variables*
 (X_ξ “does not return” to a
 recursion variable above itself)

X_σ is a *cycling variable*
 (some recursion variable below X_σ
 “returns to” X_σ)

Well-Behaved specifications (cycles back to)



$X_\sigma, X_\rho \dots$ cycling variables

X_ξ cycles back to X_σ

(The nearest return of X_ξ to a rec.var. above is to X_σ)

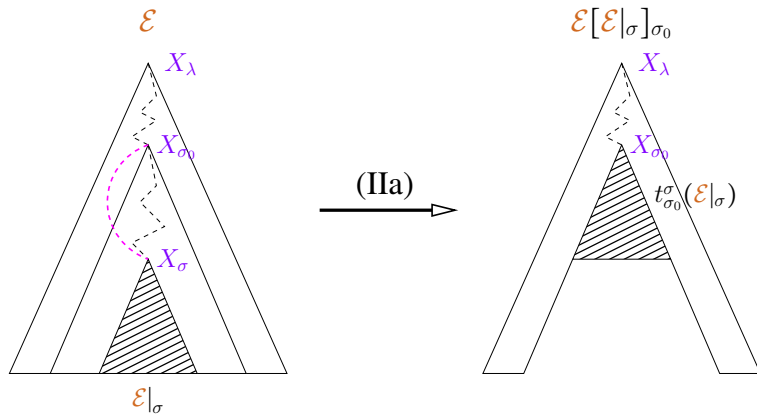
X_σ cycles back to X_ρ

$\llbracket \cdot \rrbracket_P$ -expressible = bisimilar to well-behaved

Theorem (Baeten/Corradini, 2005)

- (i) A graph G is $\llbracket \cdot \rrbracket_P$ -expressible
 $\iff G$ is the solution of a *well-behaved* specification.
- (ii) A graph G is $\llbracket \cdot \rrbracket_P$ -expressible
 $\iff$ there is a *well-behaved* 1-graph G_{wb} such that $G \leftrightarrow G_{wb}$.

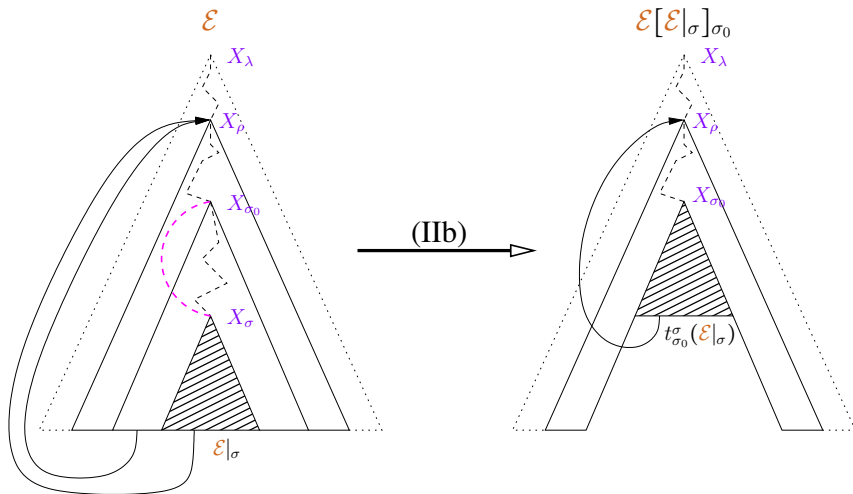
Reduction steps (IIa) for well-behaved 1-graphs



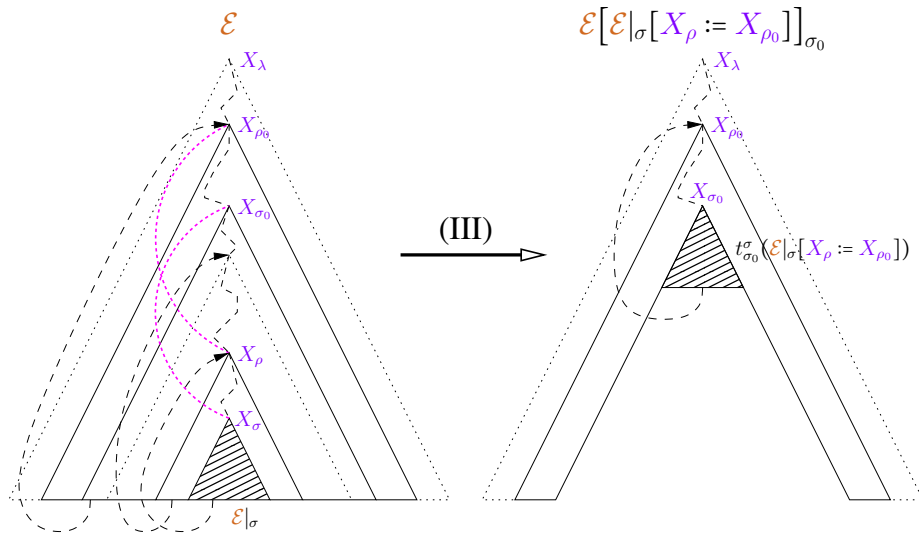
$$\langle X_\sigma | \mathcal{E} \rangle \Leftrightarrow \langle X_{\sigma_0} | \mathcal{E} \rangle$$

X_σ, X_{σ_0} are well-behaved

Reduction steps (IIb) for well-behaved 1-graphs



Reduction steps (III) for well-behaved 1-graphs



Result facilitated by steps (st1)

Theorem (*Baeten–Corradini–G, 2005*)

$\llbracket \cdot \rrbracket_P$ -Expressibility of graphs is *decidable* (at least double-exponentially).

Result facilitated by steps (st1)

Reduction Lemma for well-behaved 1-graphs (G, 2005)

Let G be graph with n states and max. out-degree k . Suppose that $\underline{G}_{wb}$ is a well-behaved 1-graph with $\underline{G}_{wb} \Leftrightarrow G$ ($\Rightarrow G, \underline{G}_{wb}$ are $\llbracket \cdot \rrbracket_P$ -expressible). Then $\underline{G}_{wb}$ can be reduced under bisimilarity by reduction steps (I)–(III) to a well-behaved 1-graph $\underline{G}_{wb}^{(0)}$ with:

- ▶ $\underline{G}_{wb}^{(0)} \Leftrightarrow \underline{G}_{wb} (\Leftrightarrow G)$,
- ▶ $\text{depth} \leq (n+1)^3 \cdot 2^{3k}$, and max. out-degree $\leq k$.

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Effective Expressibility Lemma (G, 2005)

A graph G with n states and max. out-degree k is $\llbracket \cdot \rrbracket_P$ -expressible

$$\iff G \Leftrightarrow \underline{G}_{wb}^{(0)} \text{ for a well-behaved 1-graph } \underline{G}_{wb}^{(0)} \\ \text{with } \text{depth} \leq (n+1)^3 \cdot 2^{3k} \text{ and max. out-degree } \leq k.$$

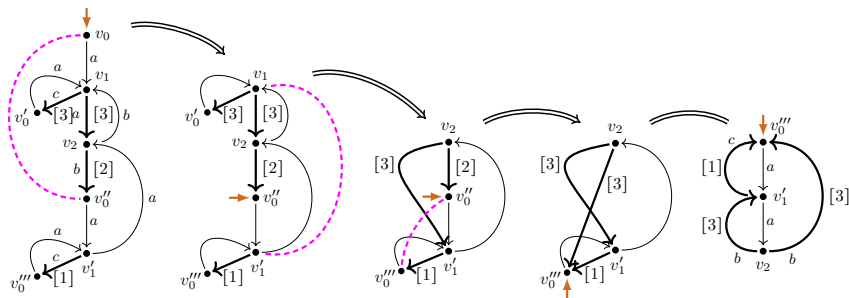
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Collapse steps for LLEE graphs under bisimilarity



Loop graphs (interpretations of innermost iterations without 1)

Definition

A process graph is a **loop graph** if:

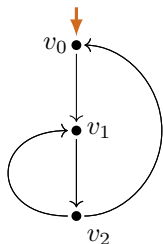
- (L1) There is an infinite path from the **start vertex**.
- (L2) Every infinite path from the **start vertex** returns to **it**.
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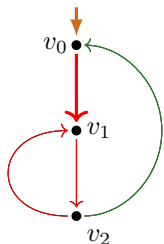


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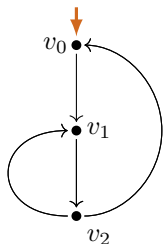
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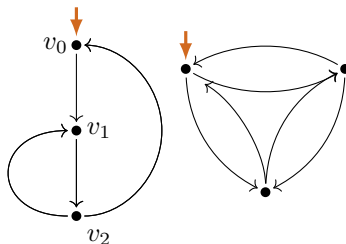
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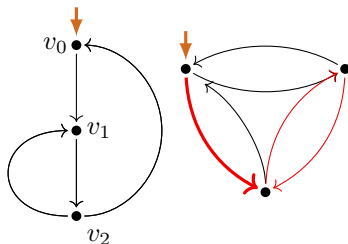
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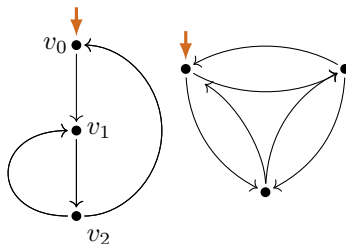
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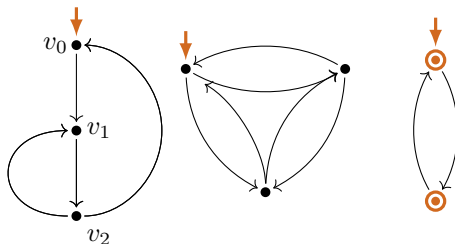
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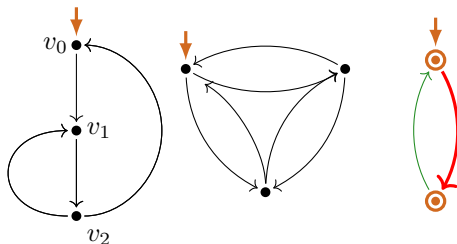
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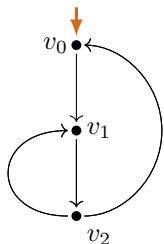
(L1), ~~(L2)~~

Loop graphs (interpretations of innermost iterations without 1)

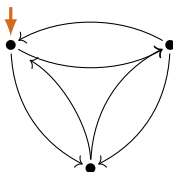
Definition

A process graph is a **loop graph** if:

- (L1) There is an infinite path from the **start vertex**.
- (L2) Every infinite path from the **start vertex** returns to it.
- (L3) Immediate termination is **only** possible at the **start vertex**.



(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~

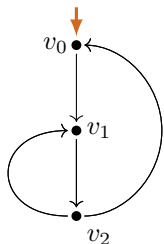


Loop graphs (interpretations of innermost iterations without 1)

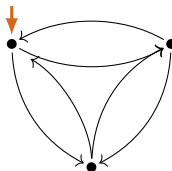
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(L1), ~~(L2)~~



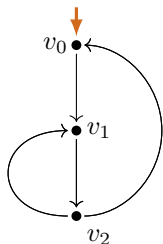
(L1), (L2), ~~(L3)~~

Loop graphs (interpretations of innermost iterations without 1)

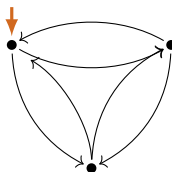
Definition

A process graph is a **loop graph** if:

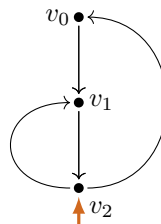
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~

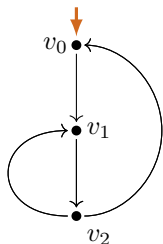


Loop graphs (interpretations of innermost iterations without 1)

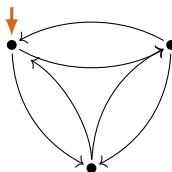
Definition

A process graph is a **loop graph** if:

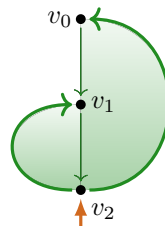
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~

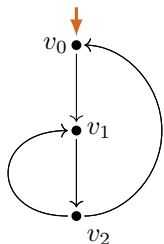


Loop graphs (interpretations of innermost iterations without 1)

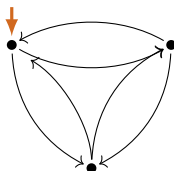
Definition

A process graph is a **loop graph** if:

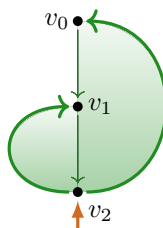
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



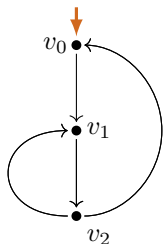
loop chart

Loop graphs (interpretations of innermost iterations without 1)

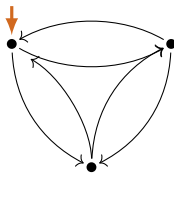
Definition

A process graph is a **loop graph** if:

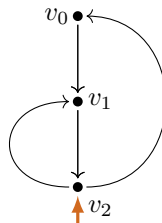
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



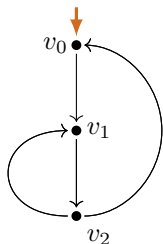
loop chart

Loop graphs (interpretations of innermost iterations without 1)

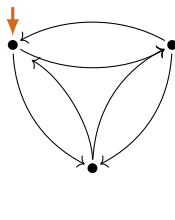
Definition

A process graph is a **loop graph** if:

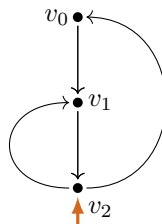
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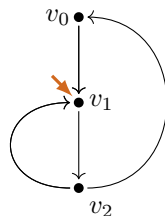
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart

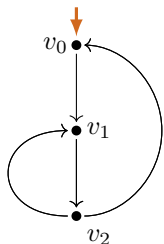


Loop graphs (interpretations of innermost iterations without 1)

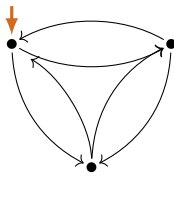
Definition

A process graph is a **loop graph** if:

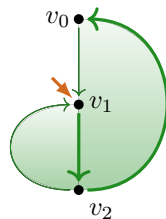
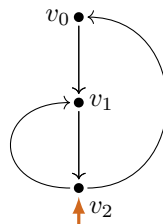
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~ loop chart

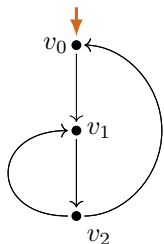


Loop graphs (interpretations of innermost iterations without 1)

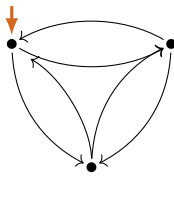
Definition

A process graph is a **loop graph** if:

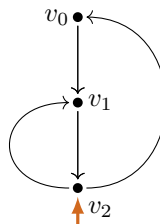
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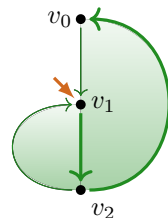
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



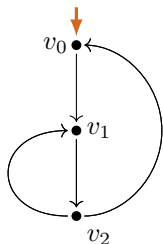
loop chart

Loop graphs (interpretations of innermost iterations without 1)

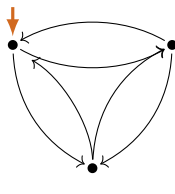
Definition

A process graph is a **loop graph** if:

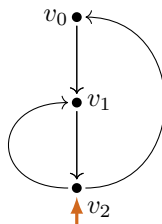
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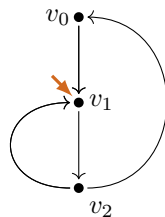
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



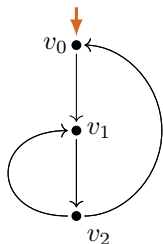
loop chart

Loop graphs (interpretations of innermost iterations without 1)

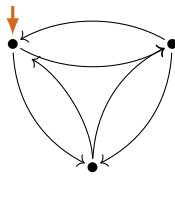
Definition

A process graph is a **loop graph** if:

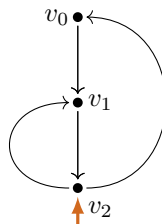
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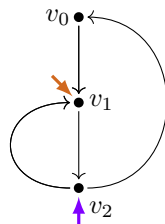
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart

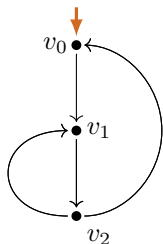


Loop graphs (interpretations of innermost iterations without 1)

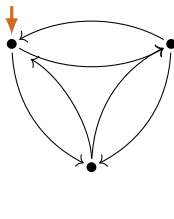
Definition

A process graph is a **loop graph** if:

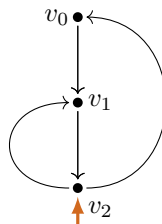
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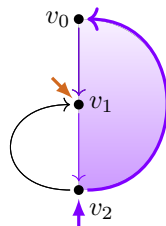
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart

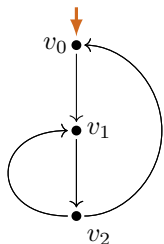


Loop graphs (interpretations of innermost iterations without 1)

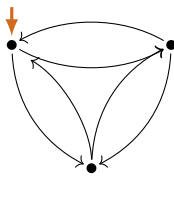
Definition

A process graph is a **loop graph** if:

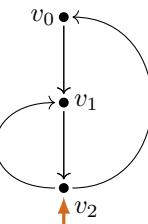
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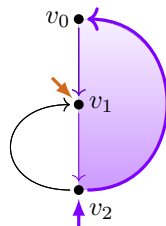
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



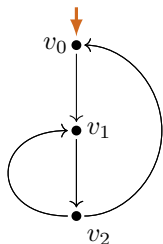
loop subchart

Loop graphs (interpretations of innermost iterations without 1)

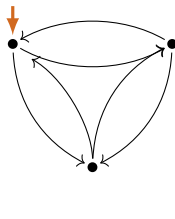
Definition

A process graph is a **loop graph** if:

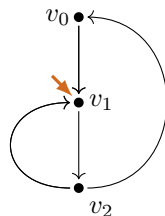
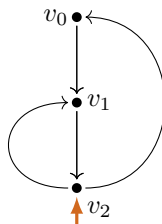
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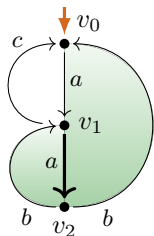
(L1), ~~(L2)~~



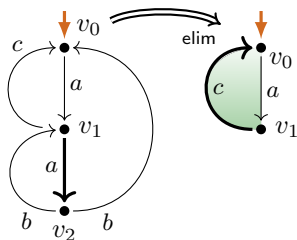
(L1), (L2), ~~(L3)~~ **loop chart**



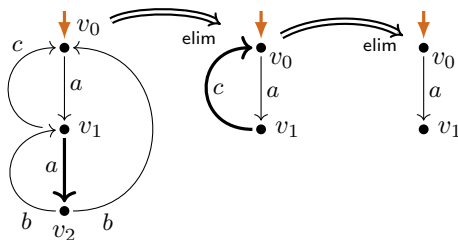
Loop existence and elimination



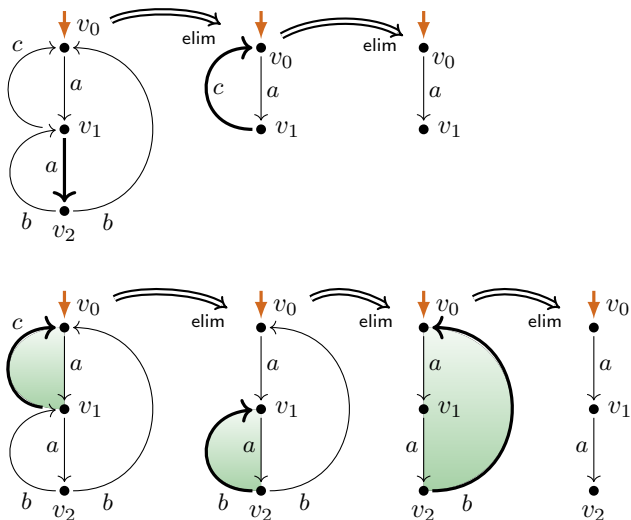
Loop existence and elimination



Loop existence and elimination



Loop existence and elimination



L

Definition

A graph G satisfies **L** (*loop existence and elimination*) if:

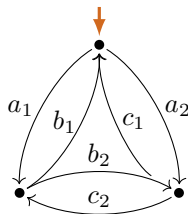
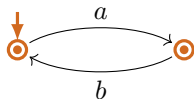
$$\exists G_0 \left(G \xrightarrow[\text{elim}]{*} G_0 \not\rightarrow_{\text{elim}} \wedge G_0 \text{ permits no infinite path} \right).$$

L

Definition

A graph G satisfies **L**EE (*loop existence and elimination*) if:

$$\exists G_0 \left(G \xrightarrow{*}_{\text{elim}} G_0 \not\rightarrow_{\text{elim}} \right. \\ \left. \wedge G_0 \text{ permits no infinite path} \right).$$

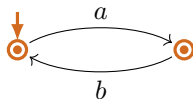


L

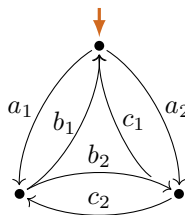
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$$\exists G_0 \left(G \xrightarrow{*}_{\text{elim}} G_0 \not\rightarrow_{\text{elim}} \right. \\ \left. \wedge G_0 \text{ permits no infinite path} \right).$$

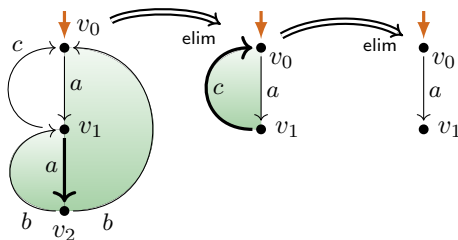


not LEE

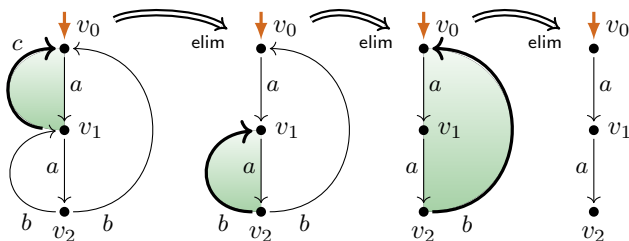
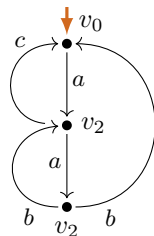


not LEE

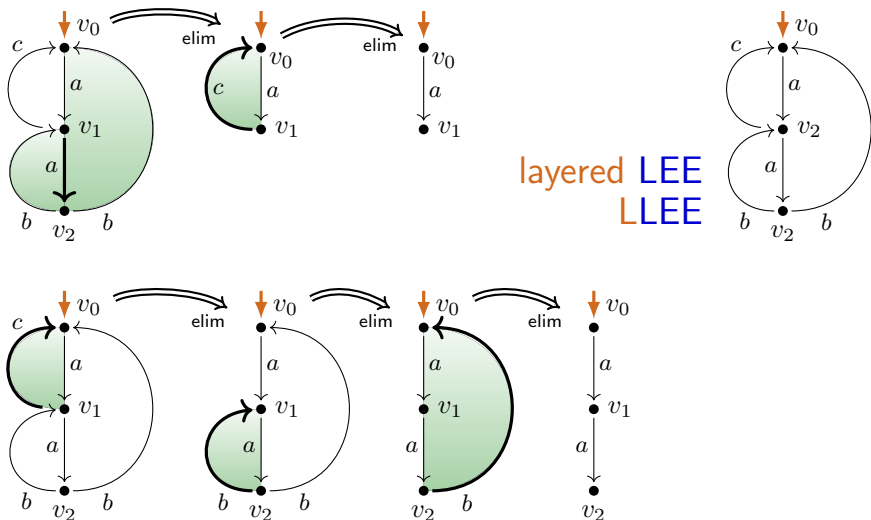
LEE



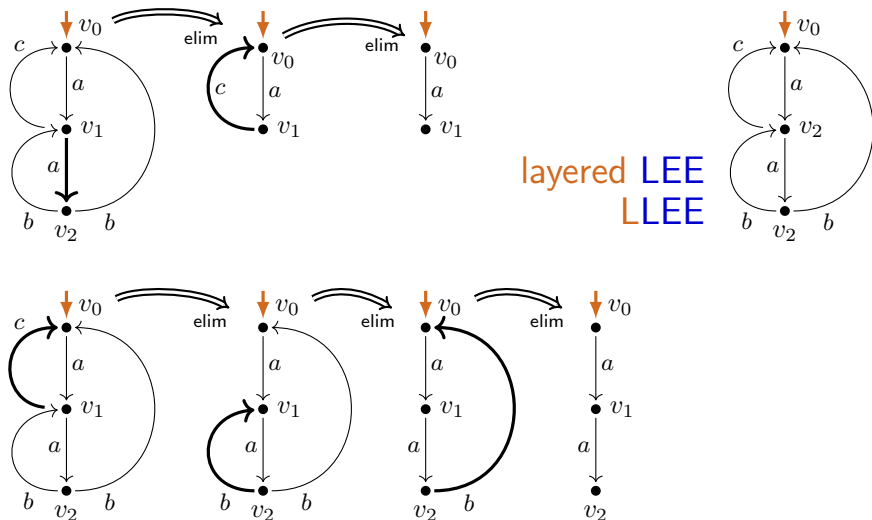
LEE



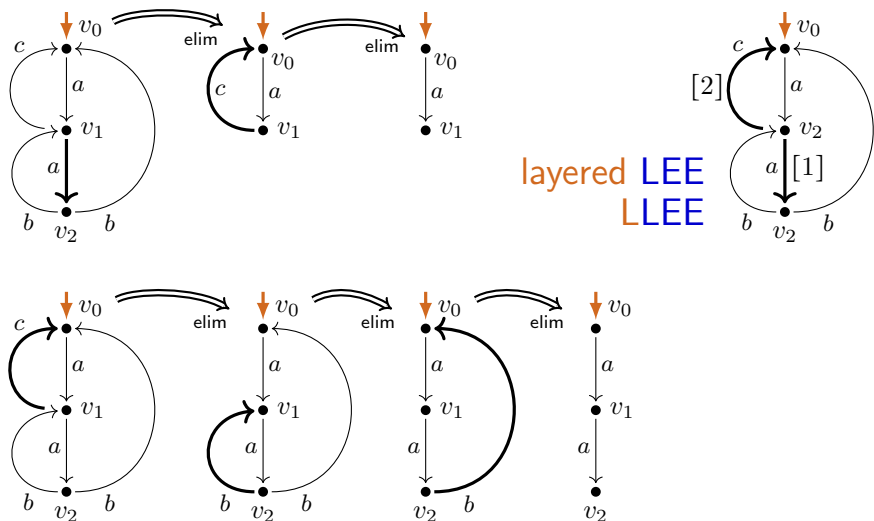
Layered LEE



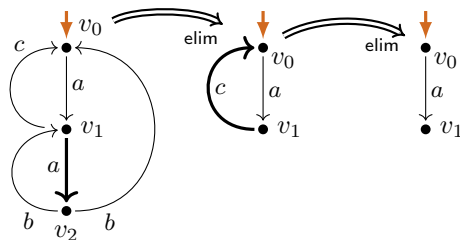
Layered LEE



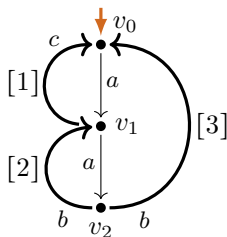
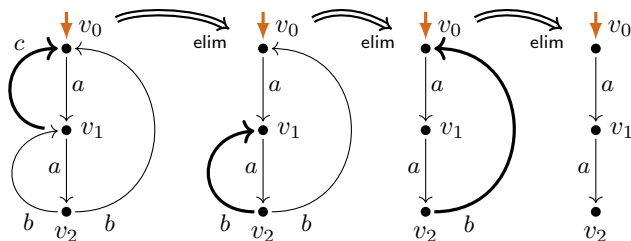
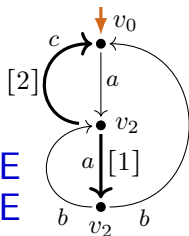
Layered LEE



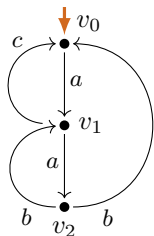
LLEE-witness and LLEE-graph



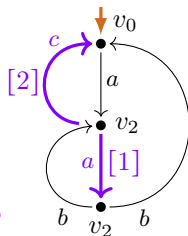
layered LEE
LLEE



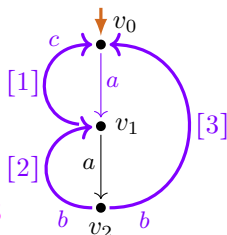
LLEE-witness and LLEE-graph



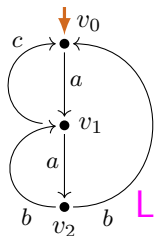
LLEE-witness



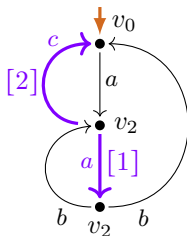
LLEE-witness



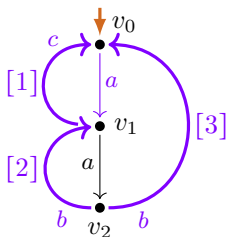
LLEE-witness and LLEE-graph



LLEE-graph

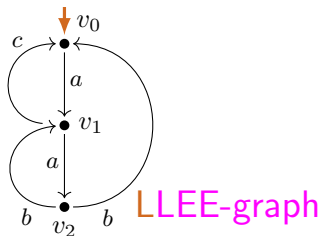


LLEE-witness

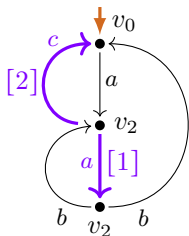


LLEE-witness

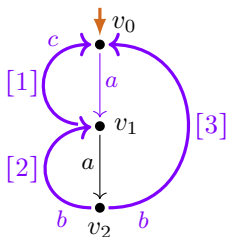
LLEE-witness and LLEE-graph



layered
LLEE-witness



layered
LLEE-witness



Properties of LEE-graphs

Theorem ($\Leftarrow$ *G-Fokkink, 2020*)

A process graph G

is $\llbracket \cdot \rrbracket_P$ -expressible by an $(*/\perp)$ regular expression

(i.e. P -expressible modulo bisimilarity by an $(*/\perp)$ reg. expr.)

if and only if

the bisimulation collapse of G satisfies LEE.

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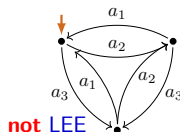
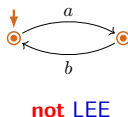
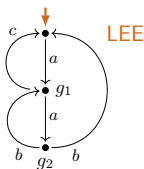
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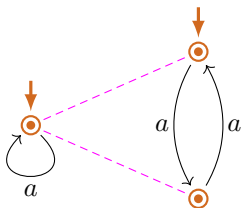
Hence $\llbracket \cdot \rrbracket_P$ -expressible | **not** $\llbracket \cdot \rrbracket_P$ -expressible by $(*/\pm)$ regular expressions:



L EE under bisimulation

Observation

- L EE is **not** invariant under bisimulation.



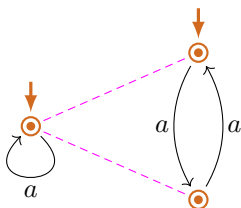
L EE

$\neg$ L EE

L EE under bisimulation

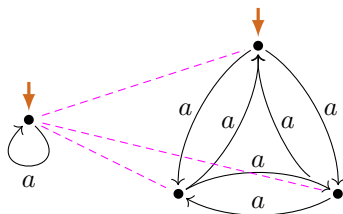
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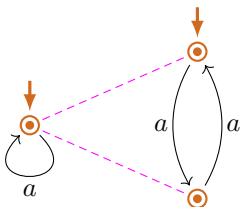
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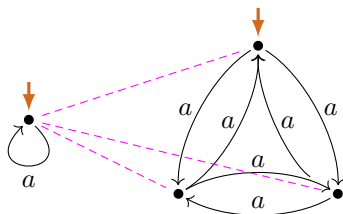
Observation

- ▶ L EE is **not** invariant under bisimulation.
- ▶ L EE is **not** preserved by converse functional bisimulation.



LEE

¬LEE



LEE

¬LEE

L_{EE} under functional bisimulation

Lemma

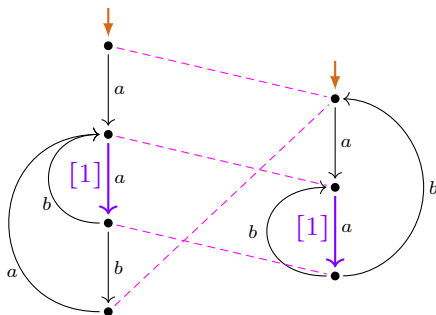
(i) **L_{EE}** is preserved by *functional bisimulations*:

$$\text{LEE}(G_1) \wedge G_1 \rhd G_2 \implies \text{LEE}(G_2).$$

Proof (Idea).

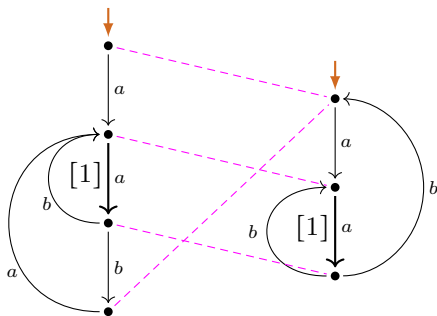
Use loop elimination in G_1 to carry out loop elimination in G_2 .

Collapsing LLEE-witnesses: global argument

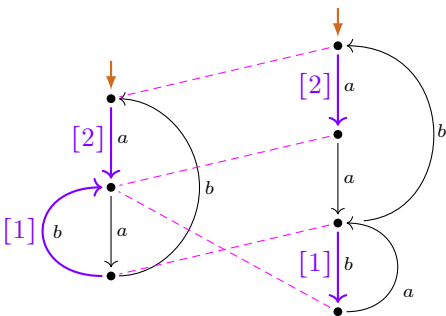


$$P(a(a(b + ba))^* \cdot 0)$$

Collapsing LLEE-witnesses: global argument



$$P(a(a(b + ba))^* \cdot 0)$$



$$P((aa(ba))^* \cdot b)^* \cdot 0)$$

L_{EE} under functional bisimulation / bisimulation collapse

Lemma

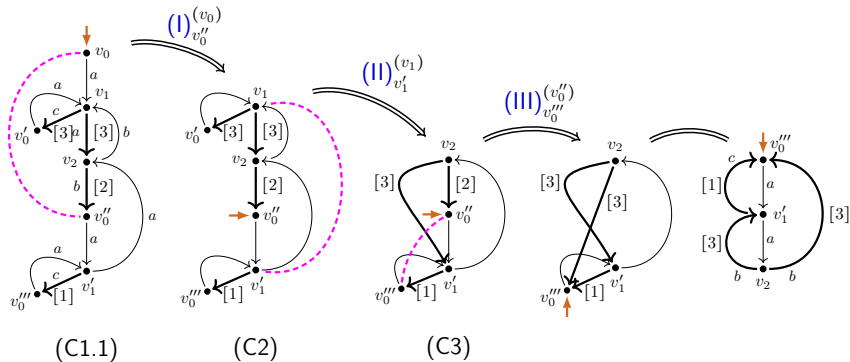
(i) **L_{EE}** is preserved by *functional bisimulations*:

$$\text{LEE}(G_1) \wedge G_1 \rightrightarrows G_2 \implies \text{LEE}(G_2).$$

(ii) **L_{EE}** is preserved from a process graph to its *bisimulation collapse*:

$$\text{LEE}(G) \wedge G \text{ has bisimulation collapse } C \implies \text{LEE}(C).$$

LLEE-preserving collapse of LLEE-graphs: stepwise proof (no 1-transitions!)

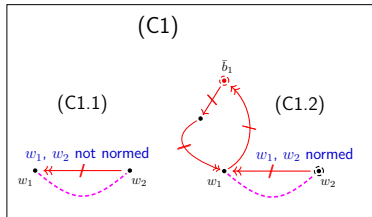


Lemma

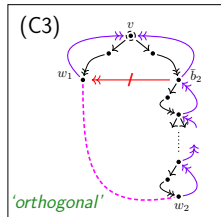
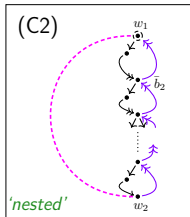
The bisimulation collapse of a LLEE-graph is again a LLEE-graph.

Reduced bisimilarity redundancies in **LLEE-graphs** (no 1-trans.!) (G-Fokkink, LICS'20)

w_1, w_2 in **different** scc's



w_1, w_2 in the **same** scc

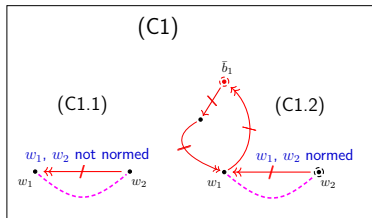


Lemma

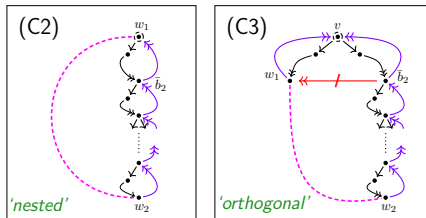
Every **not collapsed** **LLEE-graph** contains bisimilar vertices $w_1 \neq w_2$ of kind (C1), (C2), or (C3) (a **reduced bisimilarity redundancy** $\langle w_1, w_2 \rangle$):

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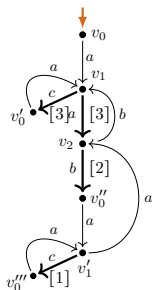
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Lemma

Every *reduced bisimilarity redundancy* in a LLEE-graph can be eliminated LLEE-preservingly.

LLEE-preserving collapse of LLEE-graphs: stepwise proof (no 1-transitions!)



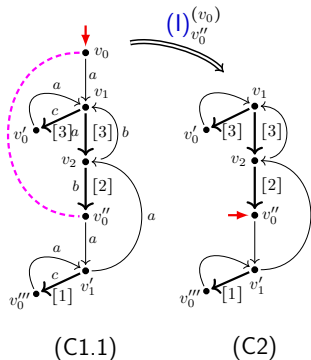
(C1.1)

Lemma

The bisimulation collapse of a LLEE-graph is again a LLEE-graph.

100%

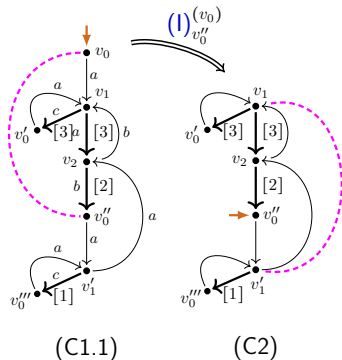
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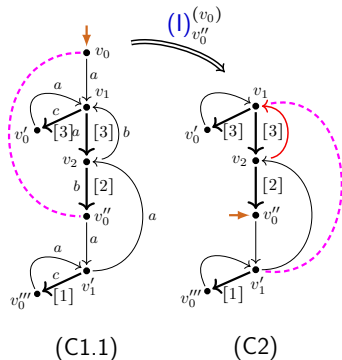
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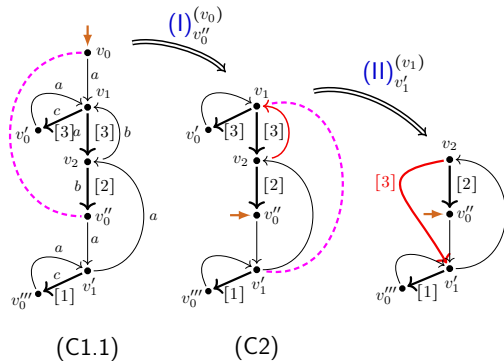
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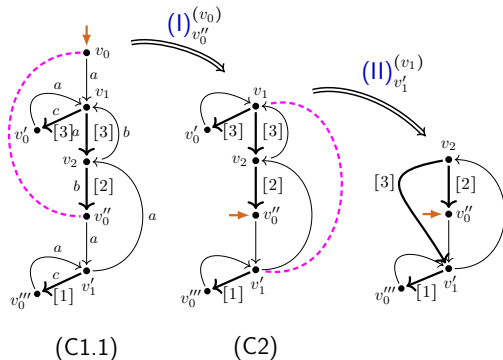
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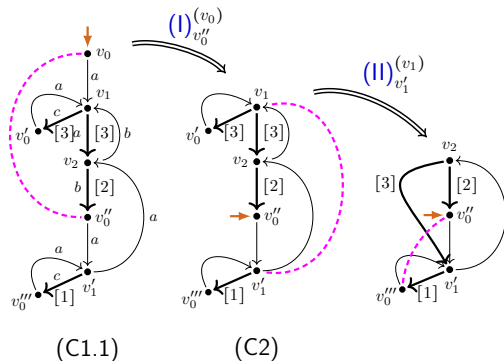
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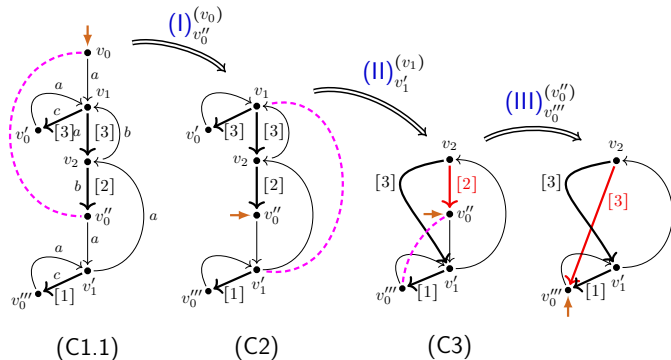
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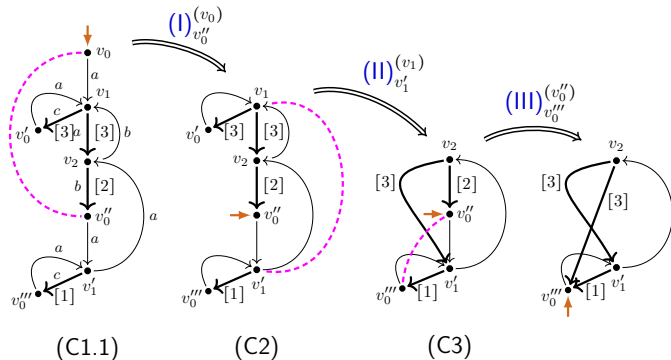
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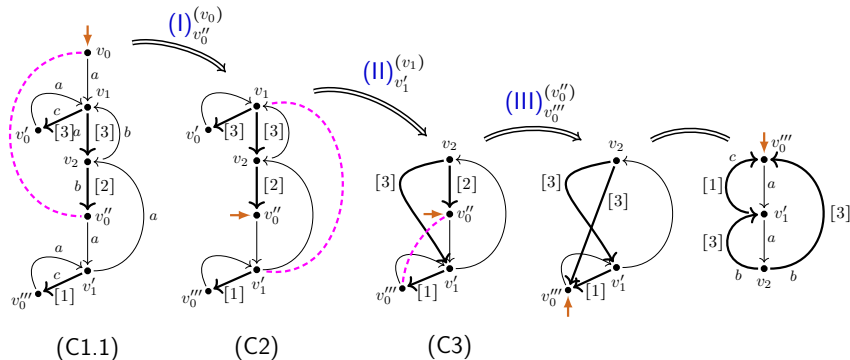
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Results facilitated by steps (st2)

Corollary ($\Leftarrow$ *G-Fokkink, LICS 2020*)

For every process graph G , TFAE:

- (i) G is $\llbracket \cdot \rrbracket_P$ -expressible by a $(*/\perp)$ regular expression
- (ii) The bisimulation collapse of G satisfies LLEE.
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Lemma (*IWC 2026*)

LLEE, LLEE $\in$ PTIME ($\leq$ cubic in graph size).

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Theorem (*G-Fokkink, LICS 2020*)

Mil restricted to $(*/\perp)$ regular expressions is complete.

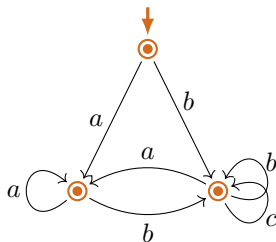
BBP (Bergstra, Bethke, Ponse)

Small steps for hard questions

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) reduction steps for well-behaved 1-graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, **as far as possible**
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback correspondences
 - (irc) LLEE/1-LLEE graphs $\hat{=}$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

graph

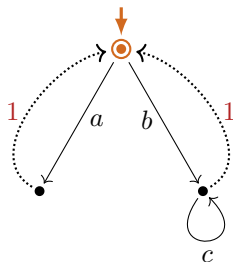


not LLEE

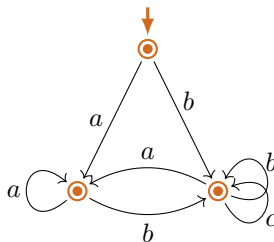
$$P((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$

1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

1-graph



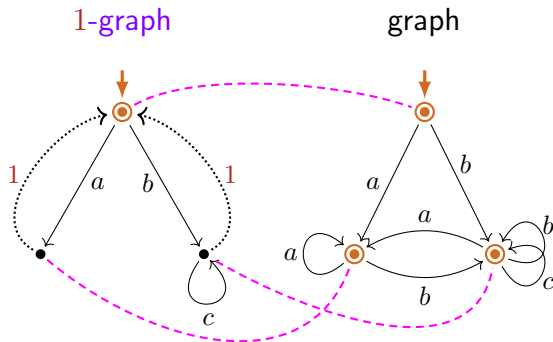
graph



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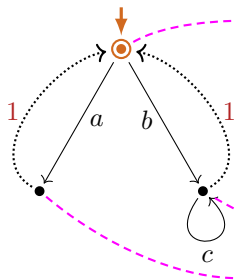
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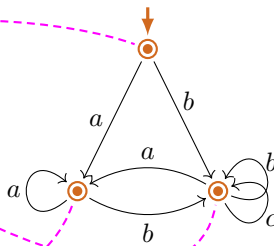
$$\underline{P}((a \cdot 1 + b \cdot (c^* \cdot 1))^*) \quad P((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$

1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

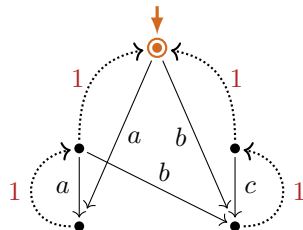
1-graph



graph



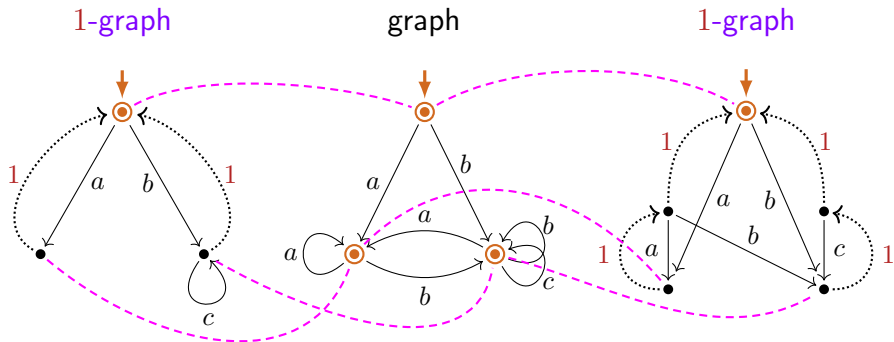
1-graph



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$$\underline{P}((a \cdot (1 \cdot ((a \cdot 1)^* \cdot 1)) \cdot b \cdot ((1 \cdot (c \cdot 1)^*)) \cdot 1 + b \cdot ((1 \cdot (c \cdot 1)^*)) \cdot 1)^*)$$

1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

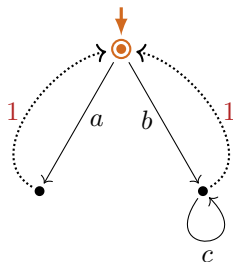


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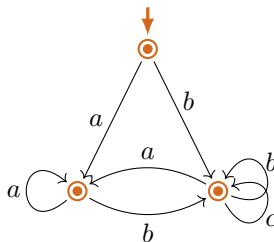
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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

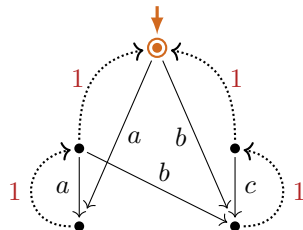
1-graph



graph



1-graph



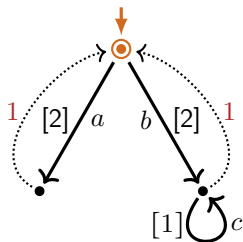
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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

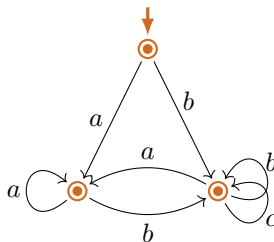
1-graph



LLEE

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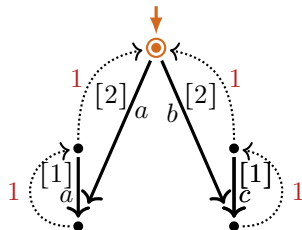
graph



not LLEE

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1-graph

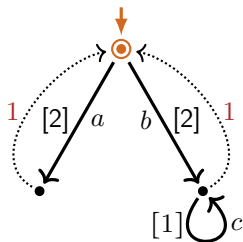


LLEE

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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

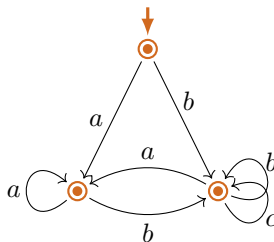
1-graph



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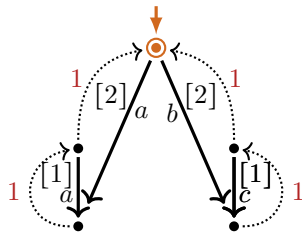
graph



1-LLEE

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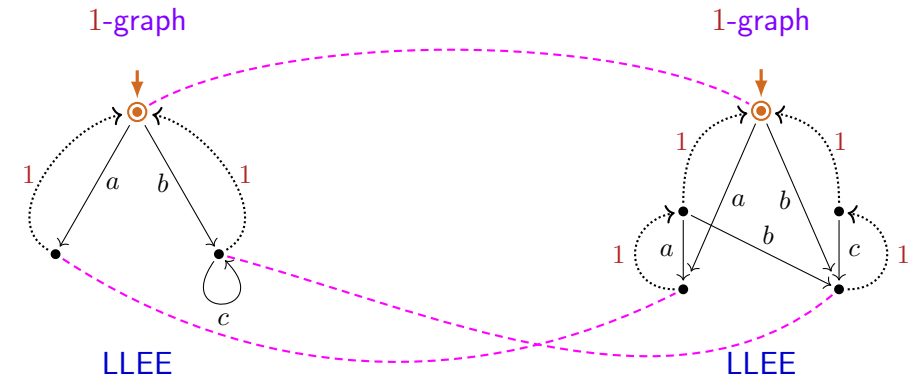
1-graph



LLEE

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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

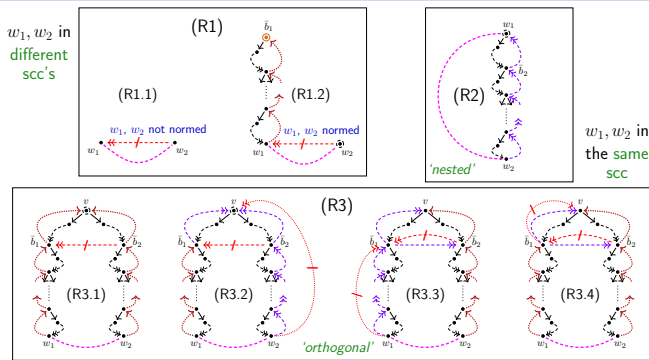


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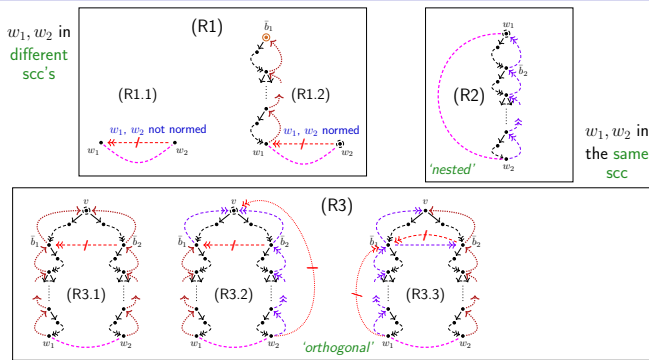
Reduced 1-bisimilarity redundancies in LLEE-1-graphs



Lemma

Every *not collapsed* LLEE-1-graph contains 1-bisimilar vertices $w_1 \neq w_2$ (a *reduced 1-bisimilarity redundancy* $\langle w_1, w_2 \rangle$) of kind (R1), (R2), (R3).

Reduced 1-bisimilarity redundancies in LLEE-1-graphs



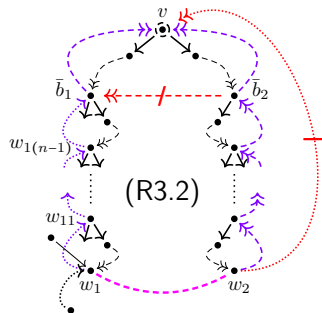
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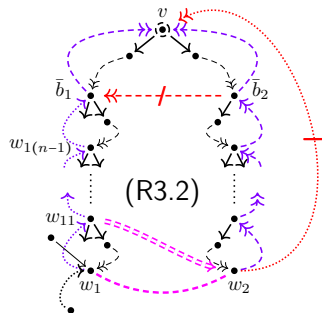
Lemma

Every *simple* reduced 1-bisimilarity redundancies in a LLEE-1-graph can be eliminated LLEE-preservingly.

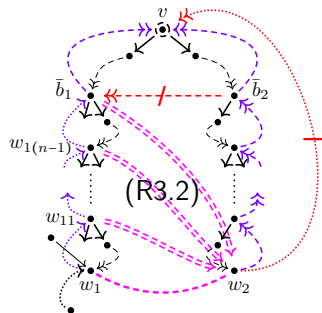
Elimination step of bisimilarity redundancies (R3.2)



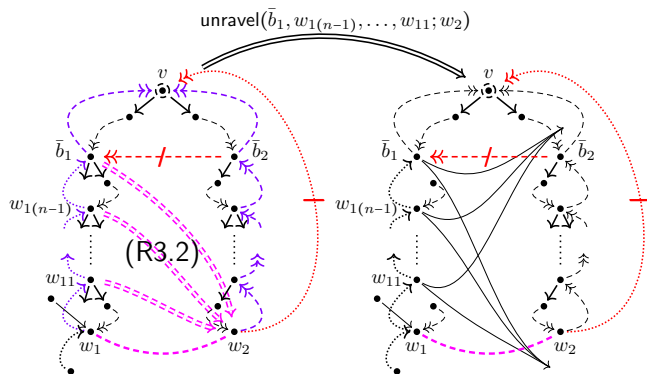
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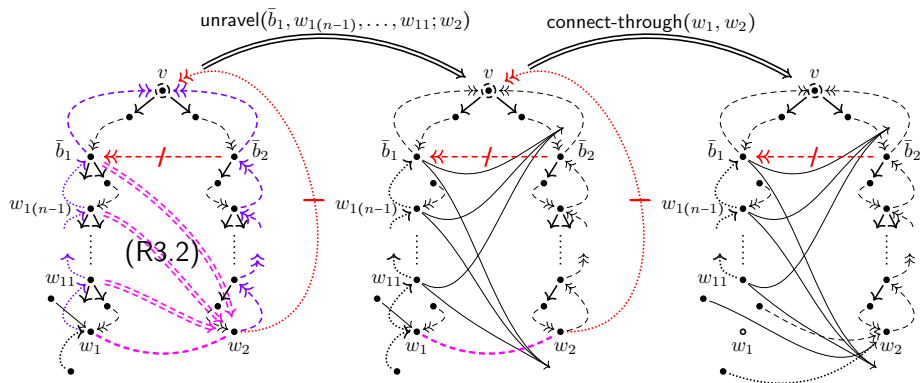
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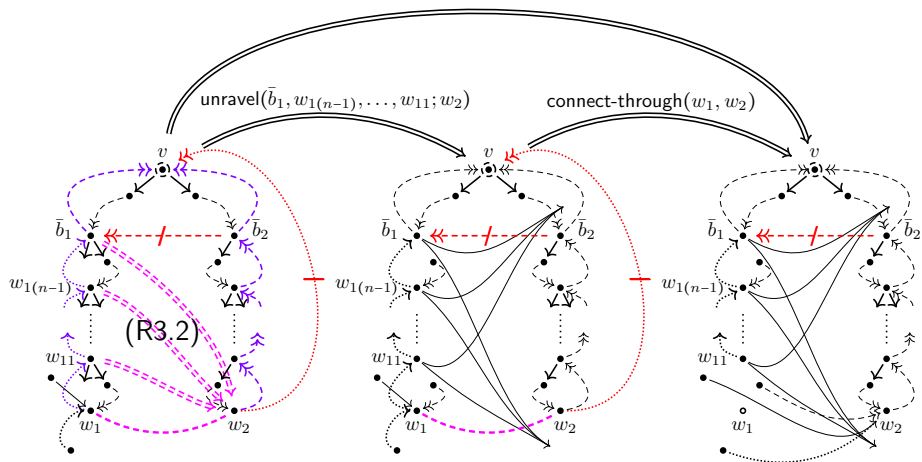
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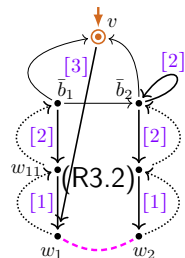
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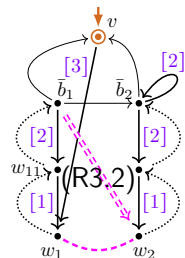
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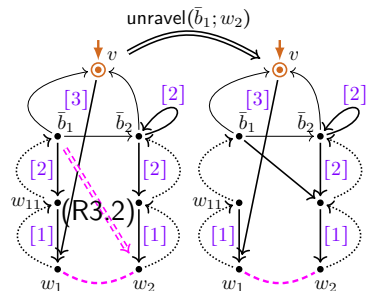
Elimination step of bisim. redundancy (R3.2) (example)



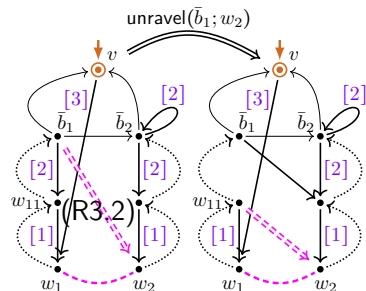
Elimination step of bisim. redundancy (R3.2) (example)



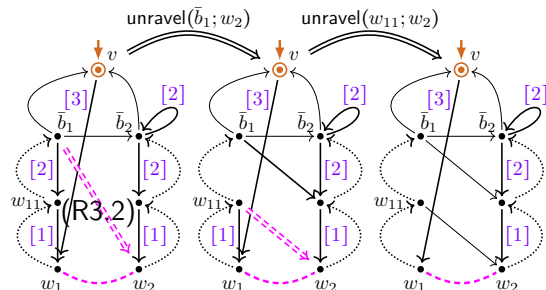
Elimination step of bisim. redundancy (R3.2) (example)



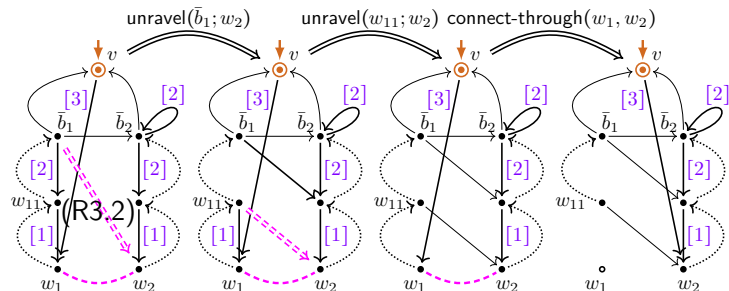
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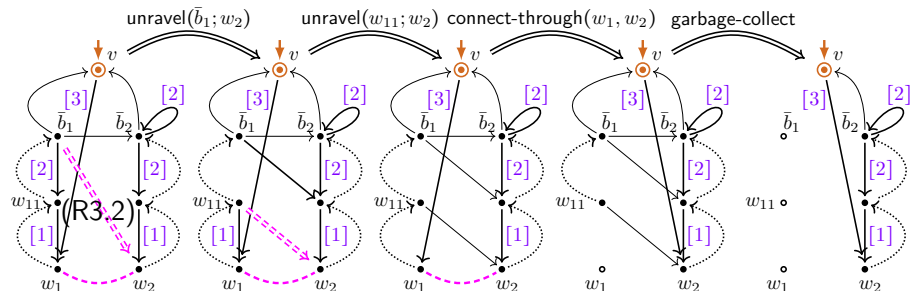
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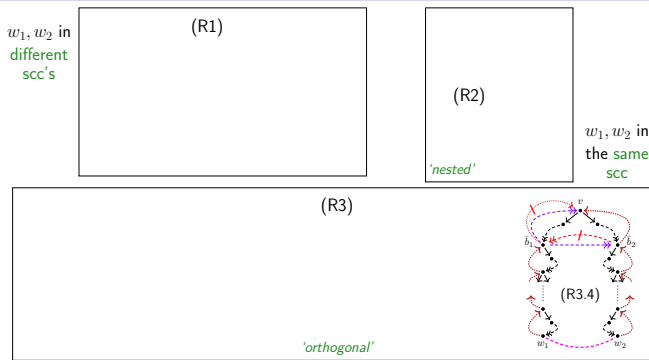
Elimination step of bisim. redundancy (R3.2) (example)



Elimination step of bisim. redundancy (R3.2) (example)



Reduced 1-bisimilarity redundancies in LLEE-1-graphs



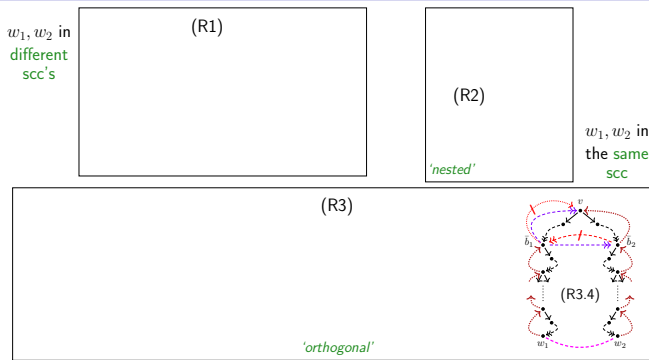
Lemma

Every *not collapsed* LLEE-1-graph contains 1-bisimilar vertices $w_1 \neq w_2$ (a *reduced 1-bisimilarity redundancy* $\langle w_1, w_2 \rangle$) of kind (R1), (R2), (R3).

Stumbling Block

How to LLEE-preservingly eliminate reduced 1-bisimilarity redundancies of kind (R3.4)?

Reduced 1-bisimilarity redundancies in LLEE-1-graphs



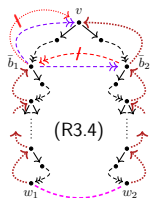
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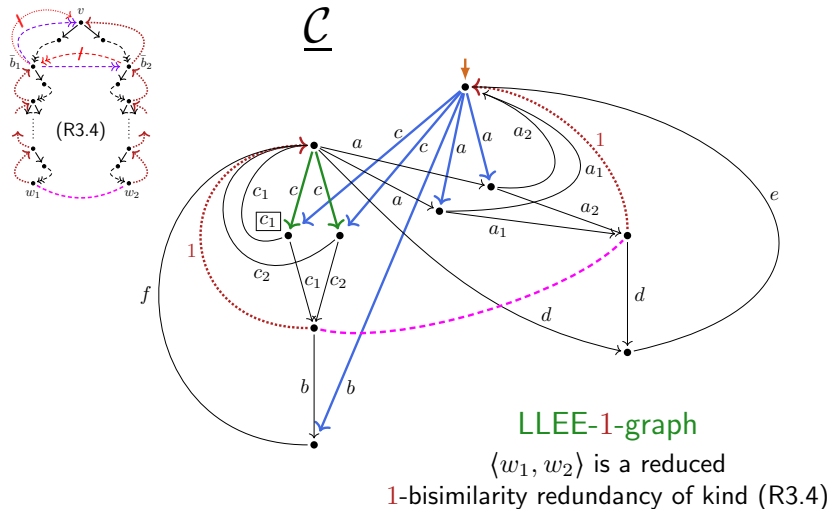
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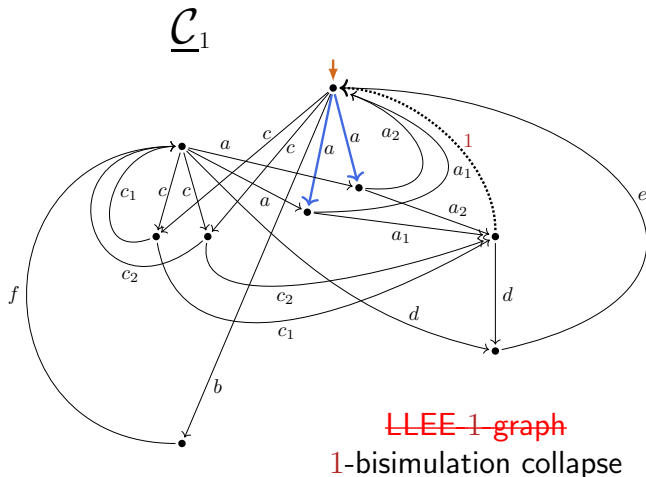
Counterexample LLEE-preserving collapse



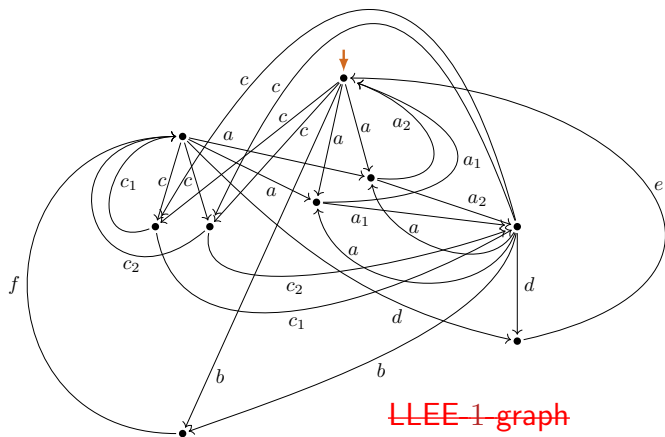
Counterexample LLEE-preserving collapse



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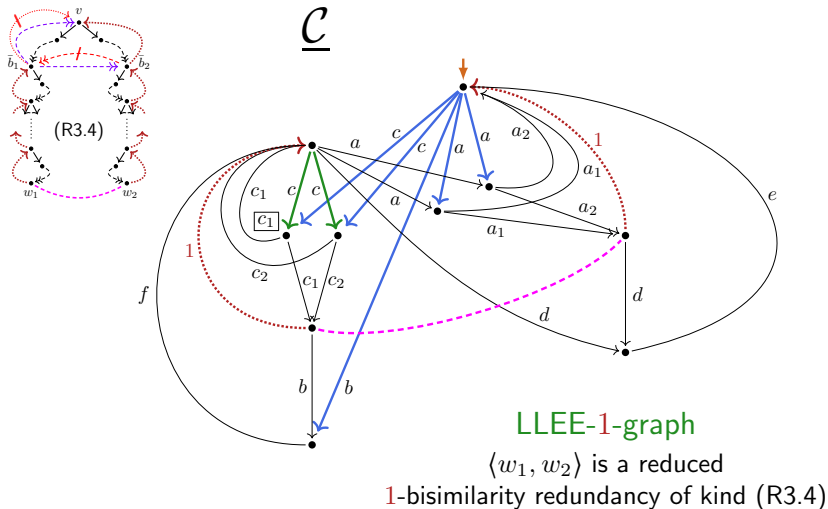


Counterexample LLEE-preserving collapse



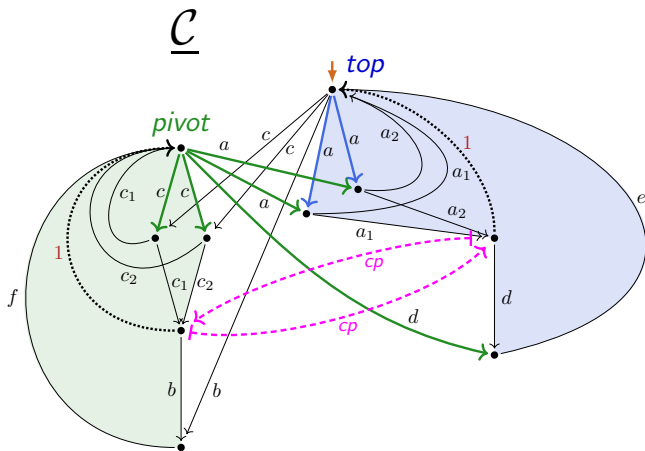
~~LLEE 1-graph~~
bisimulation collapse

Counterexample LLEE-preserving collapse

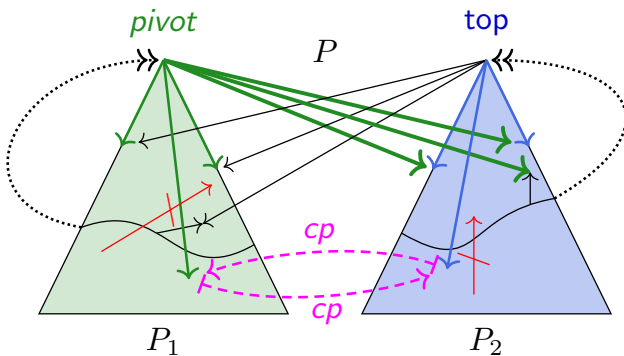




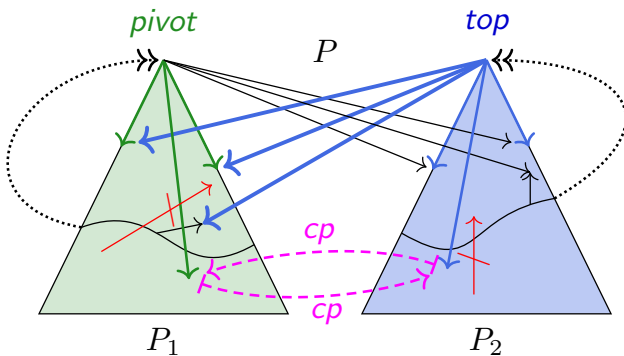
Twin-Crystal



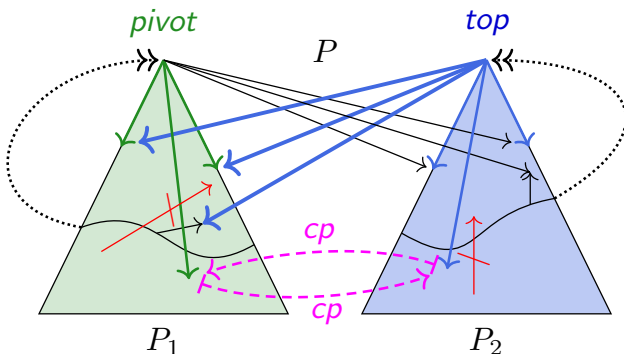
Twin-Crystal



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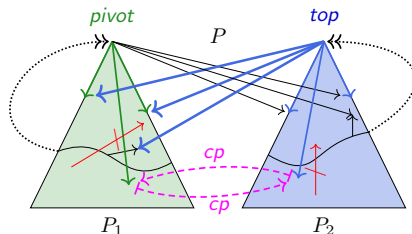


Twin-Crystal



- Provable solutions of **twin-crystals** are **complete**:
they can be transferred to their **bisimulation collapses**

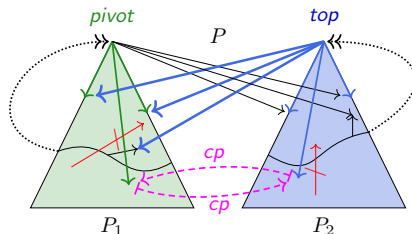
Crystallization



twin-crystal

Crystallized 1-graphs = LLEE-1-graphs that are collapsed
apart from some scc's that are twin-crystals.

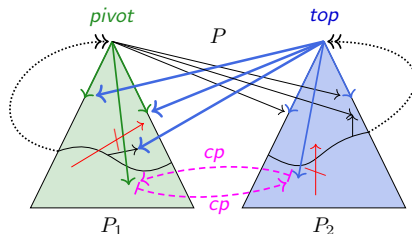
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(CR) Crystallization: Every LLEE-1-graph can be reduced under bisimilarity to a bisimilar crystallized 1-graph.

Crystallization



twin-crystal

Crystallized 1-graphs = **LLEE-1-graphs** that are collapsed apart from some scc's that are **twin-crystals**.

(CR) Crystallization: Every **LLEE-1-graph** can be reduced under **bisimilarity** to a bisimilar **crystallized 1-graph**.

(CC) Every provable solution of a **crystallized 1-graph** gives rise to provable solution on the **bisimulation collapse**.

Results facilitated by steps (st3)

Proposition

The property 1-LLEE is not preserved under bisimulation collapse.

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Crystallisation Lemma for LLEE-1-graphs

Every 1-graph $\underline{G}$ with LLEE can be pseudo-collapsed/crystallized by elimination steps for bisimilarity redundancies of kinds (R1), (R2), (R3.1), (R3.2), (R3.3) (and related steps) to a 1-graph $\underline{G}_0$ such that:

- ▶ $\underline{G}_0 \leftrightarrow \underline{G}$,
- ▶ $\underline{G}_0$ is crystallised: all strongly connected components are collapsed or twin-crystals.

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Crystallisation facilitates to **adapt** the **completeness proof** for **BBP** (that uses preservation of LLEE under collapse) via a **crystallisation detour**.

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Crystallisation Lemma for LLEE-1-graphs

Every 1-graph $\underline{G}$ with LLEE can be **pseudo-collapsed/crystallized** by **elimination steps** for bisimilarity redundancies of kinds (R1), (R2), (R3.1), (R3.2), (R3.3) (and related steps) to a 1-graph $\underline{G}_0$ such that:

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Theorem (LICS 2022)

Milner's proof system **Mil** is **complete**.

Interpretation-readback correspondences

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ Small steps for hard questions
 - (st1) reduction steps for well-behaved 1-graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, as far as possible
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation-readback correspondences
 - (irc) LLEE/1-LLEE graphs $\hat{=}$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

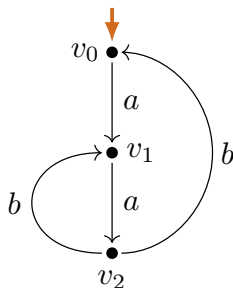
Readback

Lemma (*~ G-Fokkink, LICS 2020*)

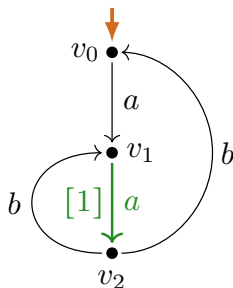
Process graphs G with LLEE are $\llbracket \cdot \rrbracket_P$ -expressible:

$$\text{LLEE}(G) \implies \exists e \in \text{RExp} \left(G \rightleftharpoons P(e) \right).$$

Readback from LLEE-witness (example)

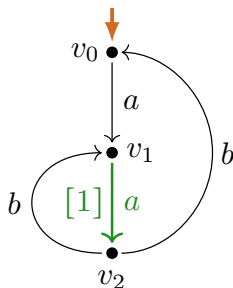


Readback from LLEE-witness (example)



layered
LEE-witness

Readback from LLEE-witness (example)



layered
LEE-witness

$$\begin{aligned} s(v_0) &= 0^* \cdot a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \end{aligned}$$

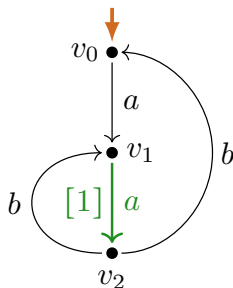
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Readback from layered LEE-witness (example)

$$s(v_0) = 0^* \cdot a \cdot s(v_1)$$

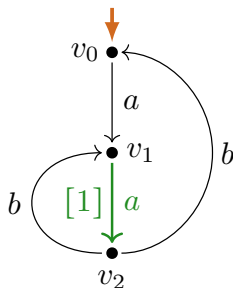


layered
LEE-witness

Readback from layered LEE-witness (example)

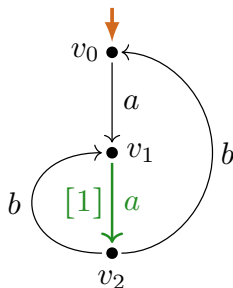
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layered
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Readback from layered LEE-witness (example)



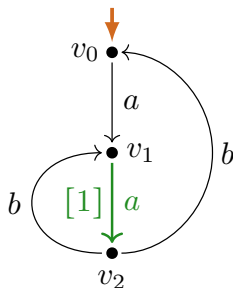
layered
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Readback from layered LEE-witness (example)



layered
LEE-witness

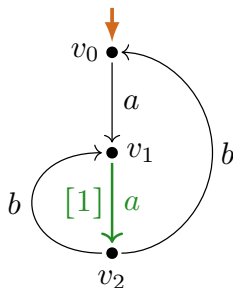
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layered
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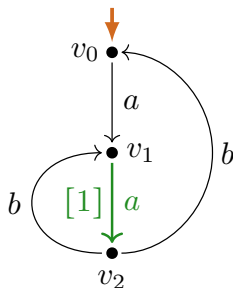
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layered
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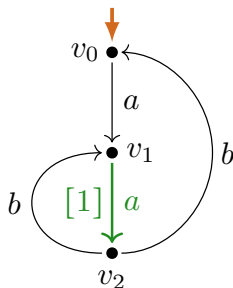
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layered
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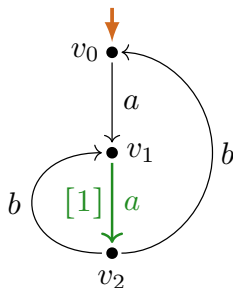
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Readback from layered LEE-witness (example)



layered
LEE-witness

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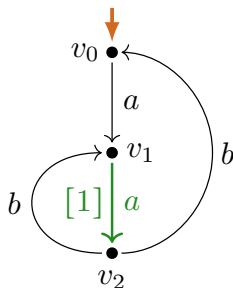
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layered
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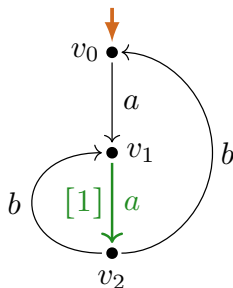
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Readback from layered LEE-witness (example)



layered
LEE-witness

$$s(v_0) = 0^* \cdot a \cdot s(v_1)$$

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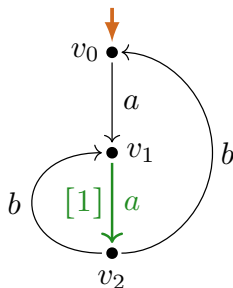
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Readback from layered LEE-witness (example)



layered
LEE-witness

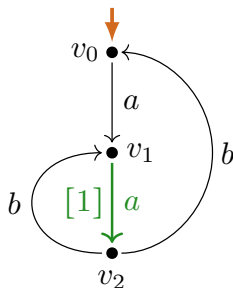
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Readback from layered LEE-witness (example)



layered
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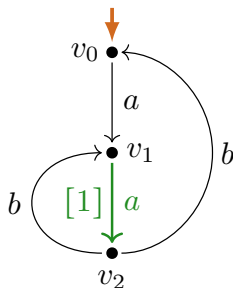
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Readback from layered LEE-witness (example)

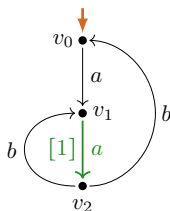


layered
LEE-witness

$$\begin{aligned}
 s(v_0) &= 0^* \cdot a \cdot s(v_1) \\
 &=_{\text{Mil}^-} a \cdot s(v_1) \\
 &=_{\text{Mil}^-} a \cdot \left(a \cdot (b + b \cdot a) \right)^* \cdot 0 \in \text{RExp}^{(*/+)} \\
 s(v_1) &= \left(a \cdot s(v_2, v_1) \right)^* \cdot 0 \\
 &=_{\text{Mil}^-} \left(a \cdot (b + b \cdot a) \right)^* \cdot 0 \\
 s(v_2, v_1) &= 0^* \cdot \left(b \cdot s(v_1, v_1) + b \cdot s(v_0, v_1) \right) \\
 &=_{\text{Mil}^-} 0^* \cdot \left(b \cdot 1 + b \cdot a \right) \\
 &=_{\text{Mil}^-} b + b \cdot a \\
 s(v_1, v_1) &= 1 \\
 s(v_0, v_1) &= 0^* \cdot a \cdot s(v_1, v_1) \\
 &= 0^* \cdot a \cdot 1 \\
 &=_{\text{Mil}^-} a
 \end{aligned}$$

Readback

Readback steps reverse P -defined transitions up to $=_{\text{Mil}^-}$ and $\leftrightarrow$.



layered
LEE-witness

$$\begin{aligned}
 s(v_0) &= 0^* \cdot a \cdot s(v_1) \\
 &=_{\text{Mil}^-} a \cdot s(v_1) \\
 &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \in \text{RExp}^{(*/+)} \\
 s(v_1) &= (a \cdot s(v_2, v_1))^* \cdot 0 \\
 &=_{\text{Mil}^-} (a \cdot (b + b \cdot a))^* \cdot 0 \\
 s(v_2, v_1) &= 0^* \cdot (b \cdot s(v_1, v_1) + b \cdot s(v_0, v_1)) \\
 &=_{\text{Mil}^-} 0^* \cdot (b \cdot 1 + b \cdot a) \\
 &=_{\text{Mil}^-} b + b \cdot a \\
 s(v_1, v_1) &= 1 \\
 s(v_0, v_1) &= 0^* \cdot a \cdot s(v_1, v_1) \\
 &= 0^* \cdot a \cdot 1 \\
 &=_{\text{Mil}^-} a
 \end{aligned}$$

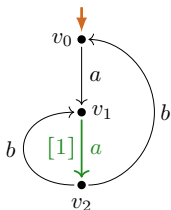
Readback

Lemma ($\sim$ G-Fokkink, 2020)

Process graphs with LLEE are $\llbracket \cdot \rrbracket_P$ -expressible by $(*/\dagger)$ reg. expressions:

$$\text{LLEE}(G) \implies \exists uf \in \text{RExp}^{(*/\dagger)} (G \Leftrightarrow P(uf)).$$

Proof: Readback steps reverse P -defined transitions up to $=_{\text{Mil}^-}$ and $\Leftrightarrow$.



layered
LEE-witness

$$\begin{aligned} s(v_0) &= 0^* \cdot a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \in \text{RExp}^{(*/\dagger)} \\ s(v_1) &= (a \cdot s(v_2, v_1))^* \cdot 0 \\ &=_{\text{Mil}^-} (a \cdot (b + b \cdot a))^* \cdot 0 \\ s(v_2, v_1) &= 0^* \cdot (b \cdot s(v_1, v_1) + b \cdot s(v_0, v_1)) \\ &=_{\text{Mil}^-} 0^* \cdot (b \cdot 1 + b \cdot a) \\ &=_{\text{Mil}^-} b + b \cdot a \\ s(v_1, v_1) &= 1 \\ s(v_0, v_1) &= 0^* \cdot a \cdot s(v_1, v_1) \\ &= 0^* \cdot a \cdot 1 \\ &=_{\text{Mil}^-} a \end{aligned}$$

Readback (refined)

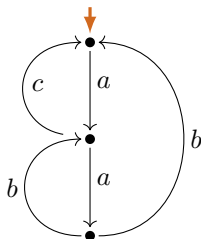
Statement

Process graphs with LLEE are P -expressible by $(*/\perp)$ reg. expressions:

$$\text{LLEE}(G) \implies \exists uf \in \text{RExp}^{(*/\perp)} (G \sim P(uf)).$$

Proof: Refine readback steps to precisely reverse P -defined transitions!

G_2



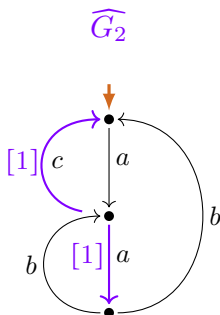
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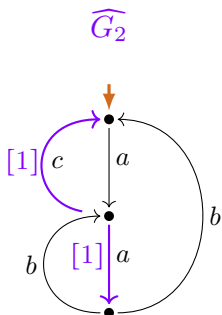
Readback (refined)

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Proof: **Refine readback steps** to **precisely reverse** P -defined transitions!



$$P((1 \cdot (\quad)^*) \cdot 0)$$

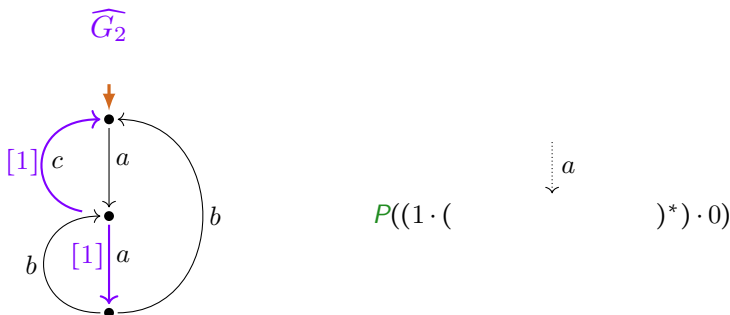
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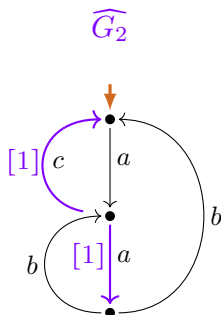
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$$P(((1 \cdot a) \cdot (\quad)^*) \cdot 0)$$

$$P((1 \cdot (\quad)^*) \cdot 0)$$

$\begin{array}{c} \vdots \\ a \\ \vee \end{array}$

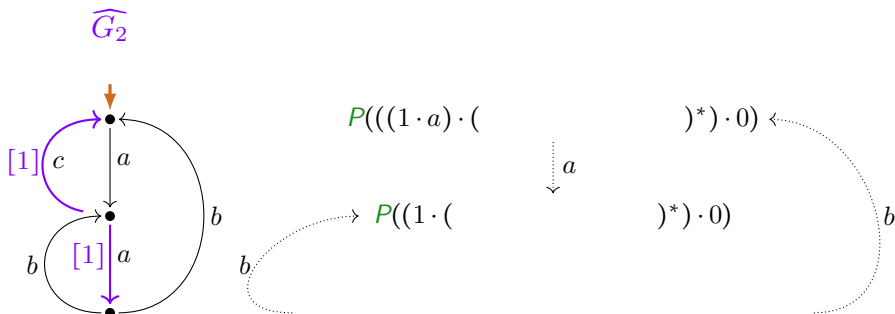
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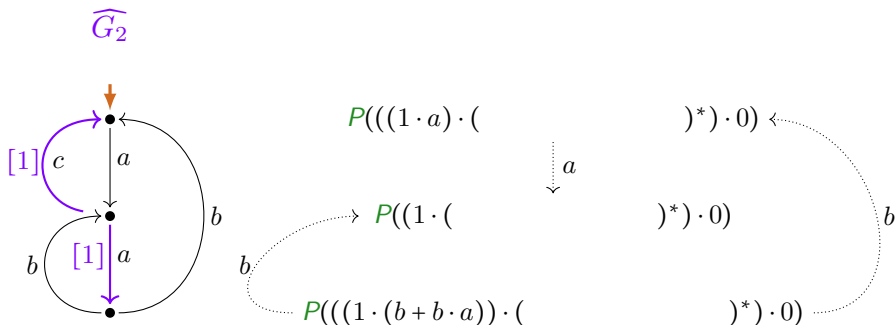
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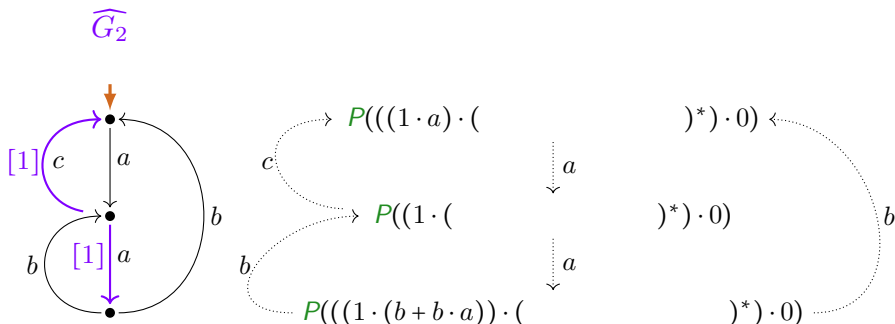
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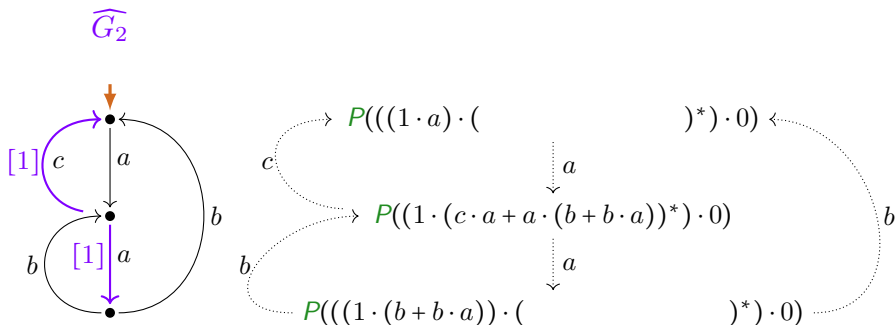
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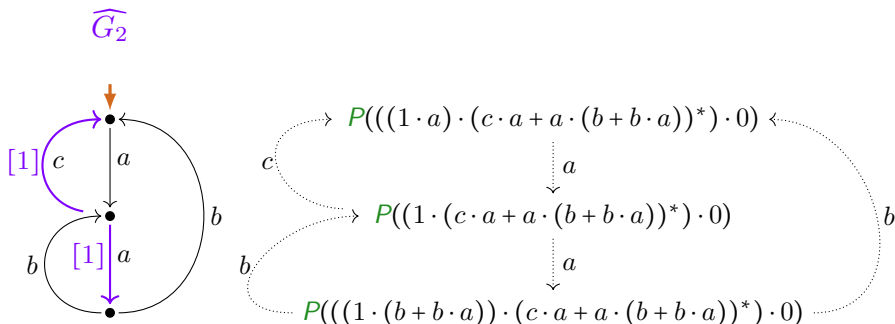
Readback (refined)

Statement

Process graphs with LLEE are P -expressible by $(*/1)$ reg. expressions:

$$\text{LLEE}(G) \implies \exists uf \in \text{RExp}^{(*/+)} (G \sim P(uf)).$$

Proof: Refine readback steps to precisely reverse P -defined transitions!



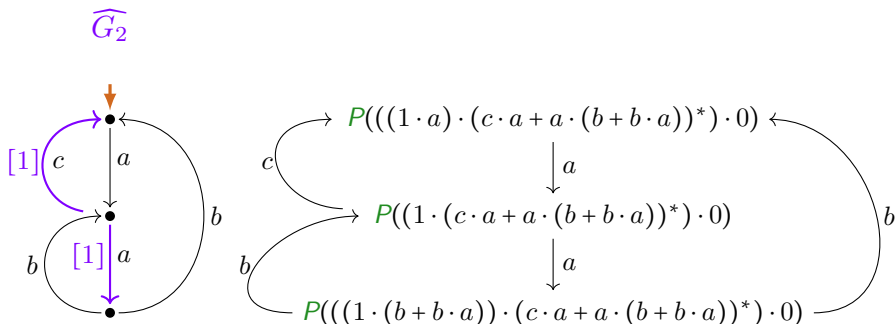
Readback (refined)

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Process graphs with **LLEE** are P -expressible by $(*/\dagger)$ reg. expressions:

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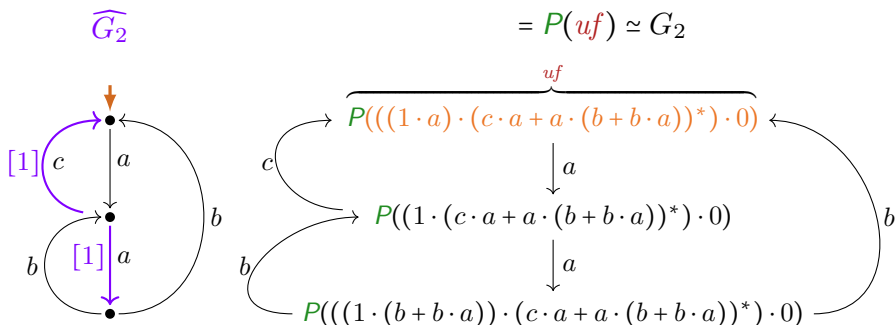
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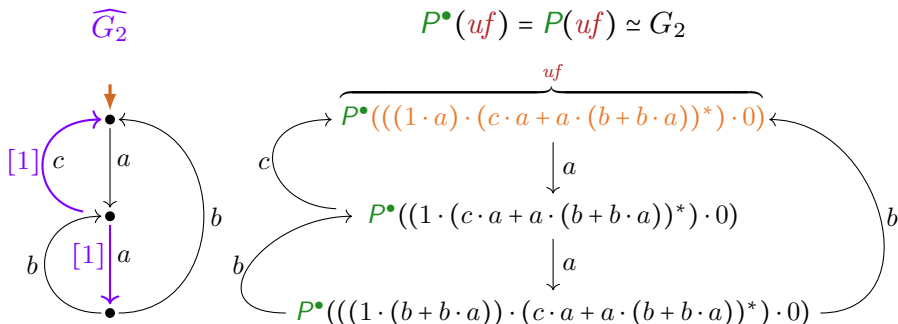
Readback (refined)

Lemma (*TERMGRAPH '24/'26*)

Process graphs with LLEE are $P^\bullet$ -expressible by $(*/1)$ reg. expressions:

$$\text{LLEE}(G) \implies \exists uf \in \text{RExp}^{(*/+)} (G \sim P^\bullet(uf)).$$

Proof: Refine readback steps to precisely reverse $P^\bullet$ -defined transitions!



Characterising $P^\bullet$ -expressibility

Theorem ($\text{LLEE} \stackrel{\wedge}{=} \text{image of } P^\bullet |_{\text{RExp}(*/\perp)}$)

A process graph G is $P^\bullet$ -expressible by an $(*/\perp)$ regular expression
 $\iff G$ is finite, and G satisfies LLEE.

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Corollary (P -expressibility by $(*/\perp)$ expr's)

For every process graph G , TFAE:

- (i) G is $[[\cdot]]_P$ -expressible by an $(*/\perp)$ regular expression.
- (ii) The bisimulation collapse of G is finite, and satisfies LLEE.
- (iii) The bisimulation collapse of G
 is $P^\bullet$ -expressible by a $(*/\perp)$ regular expression.

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Corollary (P -expressibility by $(*/\perp)$ expr's)

For every process graph G , TFAE:

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- (iii) The bisimulation collapse of G
 is $P^\bullet$ -expressible by a $(*/\perp)$ regular expression.

Theorem ($1\text{-LLEE} \stackrel{\wedge}{=} \text{image of } P^\bullet$)

A process graph G is $P^\bullet$ -expressible by a regular expression
 $\iff G$ is finite, and G satisfies 1-LLEE.

Summary

- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility question)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ Loop existence and elimination (LEE), layered form (LLEE)
- ▶ Small steps for hard questions
 - ▶ reduction steps for well-behaved specifications under bisimilarity
 - ▶ expressibility is decidable
 - ▶ collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - ▶ collapse steps, as far as possible, for 1-LLEE graphs under bisimilarity
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback correspondences
 - ▶ LEE/1-LLEE characterise restricted to $(*/\perp)$ /unrestricted image of $P^\bullet$
 - ▶ where $P^\bullet$ a sharing-increased compact refinement of P

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Next aims, and refined questions

Expressibility problems (for $P^\bullet$ and $\llbracket \cdot \rrbracket_P$)

A1: graph structure of regular expression processes (LEE/ $\mathbf{1}$ -LEE)

- ▶ $\text{LLEE} \hat{=} \text{im}(P^\bullet)((*/\mathbf{1}))$ $\mathbf{1}\text{-LLEE} \hat{=} \text{im}(P^\bullet)$ (TERMGRAPH'26)
- ▶ $\text{LLEE} \hat{=} \text{co-reducible}^+$ graphs ($\Leftarrow$ Laarakker/Kappè, TERMGRAPH'26)

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$\Leftarrow$ confluence of loop elimination with pruning steps (IWC'26),
which also entails $\text{LLEE} = \text{LEE}$.

Next aims, and refined questions

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Q1: Is 1-LLEE/ $P^\bullet$ -expressibility decidable in polynomial time?

Q2: Is $\llbracket \cdot \rrbracket_P$ -expressibility by a regular expression, for a finite process graph, decidable in polynomial time/fixed-parameter tractable time?

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Expressibility problems (for $P^\bullet$ and $[\![\cdot]\!]_P$)

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Completeness problem, solution:

A3: motivation of crystallization

A4: details of crystallization procedure,
and completeness of Milner's proof system

TERMGRAPH / IWC Talks

TERMGRAPH (tomorrow, 19 July)

CG: *Towards a Characterisation of the Graph Structure of Process Interpretations of Regular Expressions*

Ext. Abstract <https://clegra.github.io/lf/pi-struc-TG-26.pdf>

- ▶ $\text{LEE} \stackrel{\wedge}{=} \text{image of } P^\bullet |_{\text{RExp}(* / \pm)}$
- ▶ $1\text{-LEE} \stackrel{\wedge}{=} \text{image of } P^\bullet$

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IWC (Friday, 24 July)

CG: *Loop Elimination in Process Graphs is Confluent When Pruning Steps are Added*

Ext. Abstract <https://clegra.github.io/lf/le-conf-IWC-26.pdf>

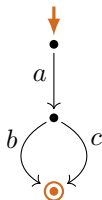
- ▶ critical-pair like analysis of loop elimination steps $\longrightarrow_{\text{elim}}$
- ▶ $\longrightarrow_{\text{elim}}$ forks are joinable by $\longrightarrow_{\text{elim}}$ steps and prune steps $\longrightarrow_{\text{prune}}$

Resources

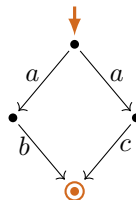
- ▶ Slides on clegra.github.io
 - ▶ slides: [.../1f/IFIP-1.6-2026.pdf](https://clegra.github.io/1f/IFIP-1.6-2026.pdf)
- ▶ CG: Closing the Image of the Process Interpretation of 1-Free Regular Expressions Under Bisimulation Collapse
 - ▶ TERMGRAPH 2024, [extended abstract](#)
- ▶ CG: The Image of the Process Interpretation of Regular Expressions is Not Closed under Bisimulation Collapse
 - ▶ [arXiv:2303.08553](#), 2021/2023.
- ▶ CG: Milner's Proof System for Regular Expressions Modulo Bisimilarity is Complete
 - ▶ LICS 2022, [arXiv:2209.12188](#), [poster](#)
- ▶ CG, Wan Fokkink: A Complete Proof System for 1-Free Regular Expressions Modulo Bisimilarity
 - ▶ LICS 2020, [arXiv:2004.12740](#), [video on youtube](#)
- ▶ CG: Modeling Terms by Graphs with Structure Constraints
 - ▶ TERMGRAPH 2018, [EPTCS 288](#), [arXiv:1902.02010](#)

Process semantics equality $=_{\llbracket \cdot \rrbracket_P}$

- Fewer identities hold for $=_{\llbracket \cdot \rrbracket_P}$ than for $=_L$:



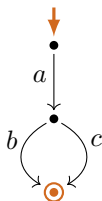
$$P(a \cdot (b + c))$$



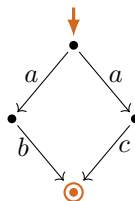
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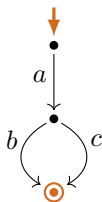
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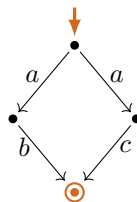
Process semantics equality $=_{[[\cdot]]_P}$

- Fewer identities hold for $=_{[[\cdot]]_P}$ than for $=_L$:



$$a \cdot (b + c)$$

$\neq_{[[\cdot]]_P}$

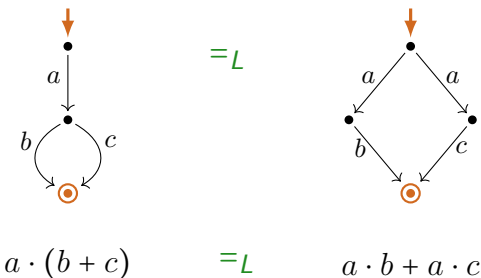


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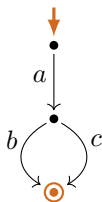
Process semantics equality $=_{\llbracket \cdot \rrbracket_P}$

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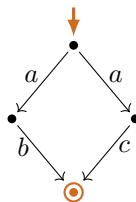
Process semantics equality $=_{\llbracket \cdot \rrbracket_P}$

- Fewer identities hold for $=_{\llbracket \cdot \rrbracket_P}$ than for $=_L$: $=_{\llbracket \cdot \rrbracket_P} \not\subseteq =_L$.



$$a \cdot (b + c)$$

$\neq_{\llbracket \cdot \rrbracket_P}$



$$a \cdot b + a \cdot c$$

$\neq_{\llbracket \cdot \rrbracket_P}$

Milner's proof system **Mil**

Axioms:

$$(A1) \quad e + (f + g) = (e + f) + g$$

$$(A2) \quad e + 0 = e$$

$$(A3) \quad e + f = f + e$$

$$(A4) \quad e + e = e$$

$$(A5) \quad e \cdot (f \cdot g) = (e \cdot f) \cdot g$$

$$(A6) \quad (e + f) \cdot g = e \cdot g + f \cdot g$$

$$(A7) \quad e = 1 \cdot e$$

$$(A8) \quad e = e \cdot 1$$

$$(A9) \quad 0 = 0 \cdot e$$

$$(A10) \quad e^* = 1 + e \cdot e^*$$

$$(A11) \quad e^* = (1 + e)^*$$

$$\text{But: } e \cdot (f + g) \neq e \cdot f + e \cdot g$$

$$\text{But: } e \cdot 0 \neq 0$$

Inference rules: rules of equational logic *plus*

$$\frac{e = f \cdot e + g}{e = f^* \cdot g} \text{RSP}^* \text{ (if } f \text{ does not terminate immediately)}$$

Milner's Completeness Question

Is **Mil** complete with respect to $=_{[\cdot]_P}$? (Does $e =_{[\cdot]_P} f \implies e =_{\text{Mil}} f$ hold?)

Milner's questions

(Q1) Completeness Question:

*Is the proof system **Mil** **complete** for $=_{\llbracket \cdot \rrbracket_P}$?*

(Q2) $\llbracket \cdot \rrbracket_P$ -Expressibility Question:

*What **structural property** characterizes
process graphs that are $\llbracket \cdot \rrbracket_P$ -**expressible** ?*

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- ▶ series of partial completeness results for:
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 - ▶ with a stronger fixed-point rule (G, 2006)
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- ▶ Yes! (G, 2022, proof summary, employing LEE and crystallization)
- ▶ series of partial completeness results for:
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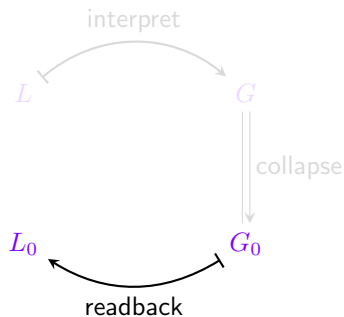
Question (Q2) specialized

(Q2)₀ P -Expressibility and P - $(*/\perp)$ -Expressibility:

What *structural property* characterizes:

- ▶ process graphs that are P -expressible ?
(... in the *image of P* ?)
- ▶ process graphs that are P -expressible by $(*/\perp)$ regular expressions?
(... in the *image of $(*/\perp)$ expressions under P* ?)

Readback



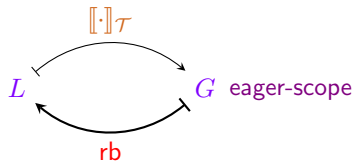
Readback

defined with property:



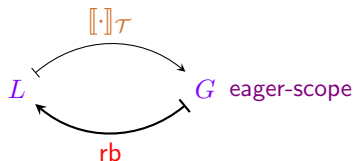
Readback

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Readback

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Theorem

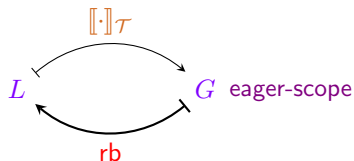
For all *eager-scope* λ -term-graphs G :

$$([[\cdot]]_{\mathcal{T}} \circ \text{rb})(G) \simeq G$$

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Readback

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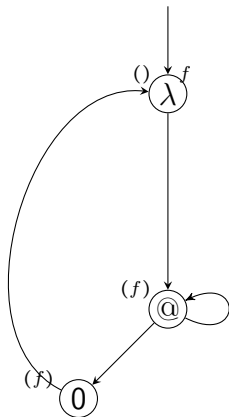
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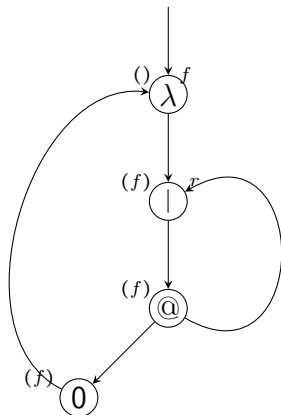
idea:

1. construct a spanning tree T of G
2. using local rules, in a bottom-up traversal of T synthesize $L = \text{rb}(G)$

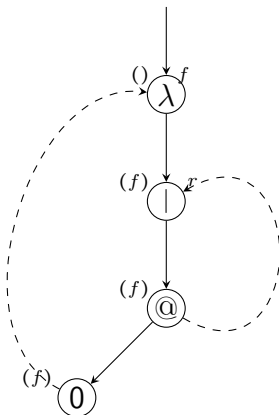
Readback: example (fix)



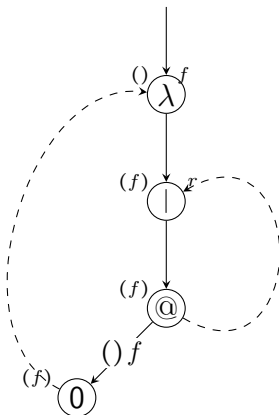
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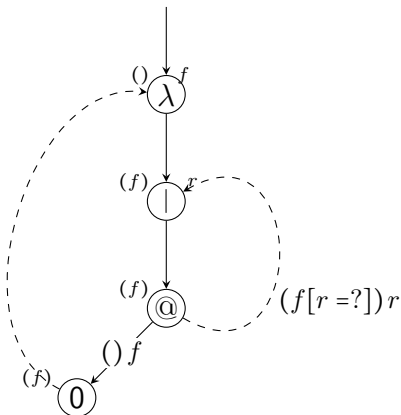
Readback: example (fix)



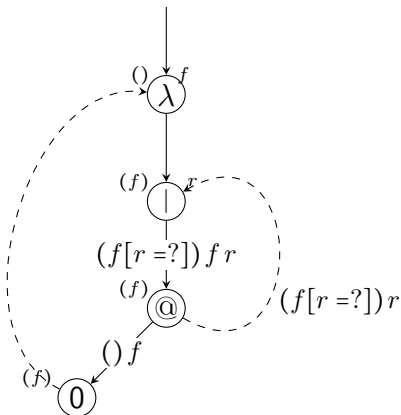
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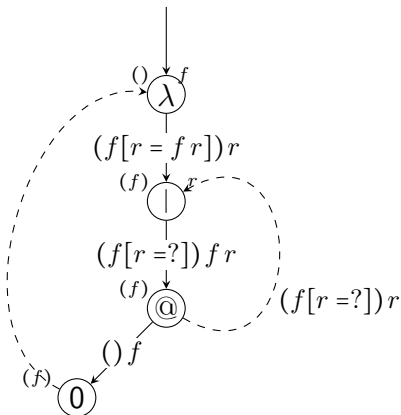
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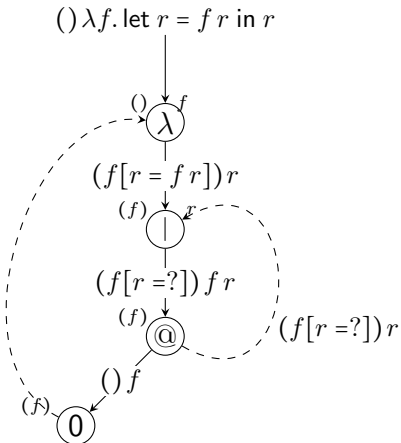
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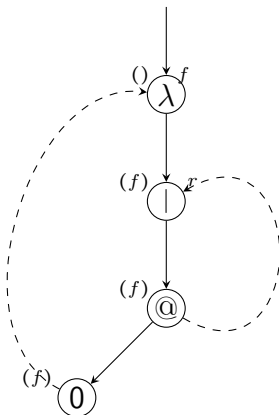
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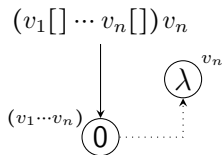
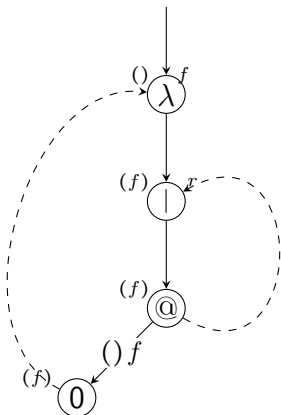
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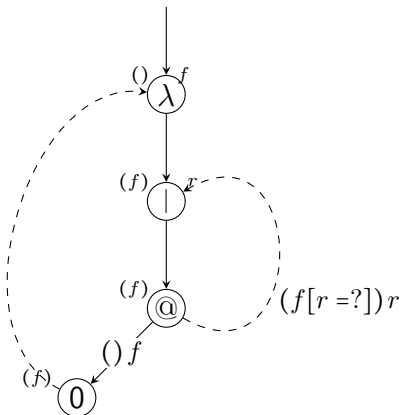
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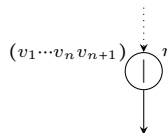
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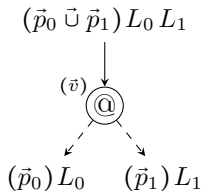
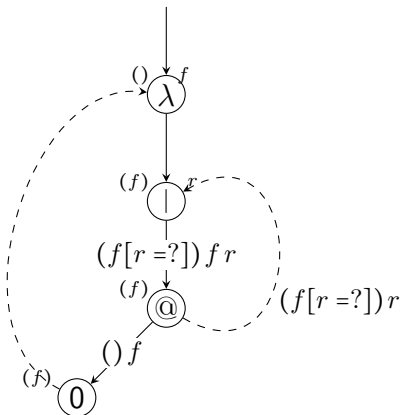
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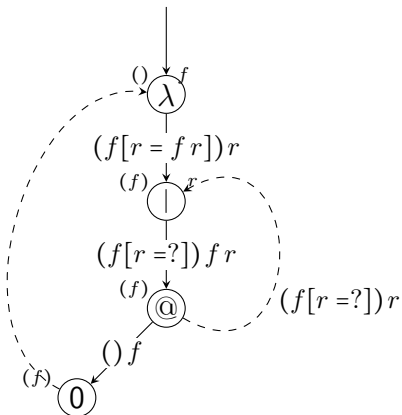
$(v_1[] \cdots v_n[] v_{n+1}[r = ?])r$



Readback: example (fix)



Readback: example (fix)



$$\begin{array}{c}
 (\vec{p} \ v_{n+1}[B, r = L])r \\
 \downarrow \\
 (vs(\vec{p}) \ v_{n+1}) \downarrow r \\
 \downarrow \\
 (\vec{p} \ v_{n+1}[B, (r = ?)])L
 \end{array}$$

Readback: example (fix)

